

Process flowsheet development

KEY LEARNING OBJECTIVES

- How to read and draw a process flow diagram (PFD)
- When to design a batch process or a continuous process
- Factors to consider when adopting or improving commercially proven technology
- How to develop a flowsheet for a revamp design
- How to synthesize a flowsheet for an entirely new process
- How to review a flowsheet and check for completeness and errors

2.1 Introduction

This chapter covers the preparation and presentation of the process flowsheet, also known as the *process flow diagram* (PFD). The flowsheet is the key document in process design. It shows the arrangement of the equipment selected to carry out the process, the stream connections, the stream flow rates and compositions, and the operating conditions. It is a diagrammatic model of the process. Chemical engineers in industry are usually very proficient at reading PFDs and use it as the primary means of transmitting and recording process information.

The flowsheet is used by specialist design groups as the basis for their designs. These include piping, instrumentation, and equipment design and plant layout. It is also used by operating personnel for the preparation of operating manuals and operator training. During plant start-up and subsequent operation, the flowsheet forms a basis for the comparison of operating performance with design. If the plant is later revamped to new specifications, the PFD of the original plant is the starting point for the revamp design.

Several types of PFDs are used by chemical engineers, depending on the level of detail required. A simple block flow diagram can be used to give a rough idea of the overall process flow structure and may be useful when giving a presentation. A full PFD should include all of the process vessels and equipment and show all the process and utility flow lines. A full heat and material balance of the process showing the composition, flow rate, and temperature of every stream is usually included in or with a PFD. The PFD also indicates the location of every control valve, as control valves play an important role in determining the pressure balance of the process and hence in the sizing of pumps and compressors. A piping and instrumentation diagram (P&ID) is a more detailed version of the PFD that also includes information on ancillary instruments and valves, sampling and drain lines, start-up and shut-down systems, and pipe sizes and metallurgy. The P&ID is used in detailed design and safety analysis.

This chapter presents an overview of how to read and draw flowsheets and discusses how the unit operations of a process are selected to form the basic process flow structure. [Chapter 3](#) discusses energy flows within a process and describes heat and power recovery methods that are used to make processes more energy-efficient. [Chapter 4](#) covers the use of commercial process simulation tools to generate the heat and material balances for the flowsheet, and [Chapter 5](#) introduces the elements of process control that must be understood to fill in the control systems on the PFD.

2.2 Flowsheet presentation

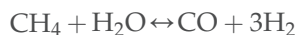
As the process flowsheet is the definitive document on the process, the presentation must be clear, comprehensive, accurate, and complete. This section describes how to read and draw PFDs.

2.2.1 Block diagrams

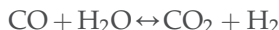
A block diagram is the simplest form of flow diagram. Each block can represent a single piece of equipment or a complete stage in the process. Block diagrams are useful for showing simple processes. For complex processes, their use is limited to showing the overall process, broken down into its principal stages.

Block diagrams are useful for representing a process in a simplified form in reports, textbooks, and presentations but have limited use as engineering documents. The stream flow rates and compositions can be shown on the diagram adjacent to the stream lines when only a small amount of information is to be shown or can be tabulated separately.

[Fig. 2.1](#) shows a block flow diagram of a steam reforming process for making hydrogen from methane. The methane feed enters on the left and is mixed with steam and preheated in the convective section of a fired heater. The steam–methane mixture then passes through the reactor tubes in the radiant section of the heater, where the steam reforming reaction takes place:



The products from the steam reformer are sent to a shift reactor. The shift reactor increases the amount of hydrogen in the product by allowing the water–gas shift reaction to re-equilibrate at a lower temperature:



The shift reactor products are then further cooled and scrubbed in an absorber to remove carbon dioxide before being sent to a pressure-swing adsorption process that separates hydrogen from carbon dioxide, unconverted methane, and water vapor.

Block diagrams are often drawn using simple graphics programs such as Microsoft Visio or PowerPoint. Block flow diagrams are not restricted to using boxes to represent process operations and often use the same flowsheet symbols used in PFDs. Generally, any flowsheet sketch that does not show all the required process equipment should be classified as a block flow diagram.

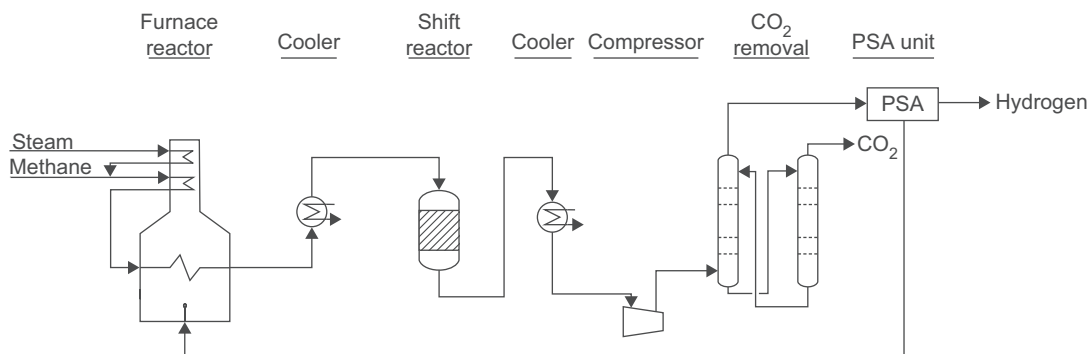


FIG. 2.1 Block flow diagram of steam reforming process for hydrogen.

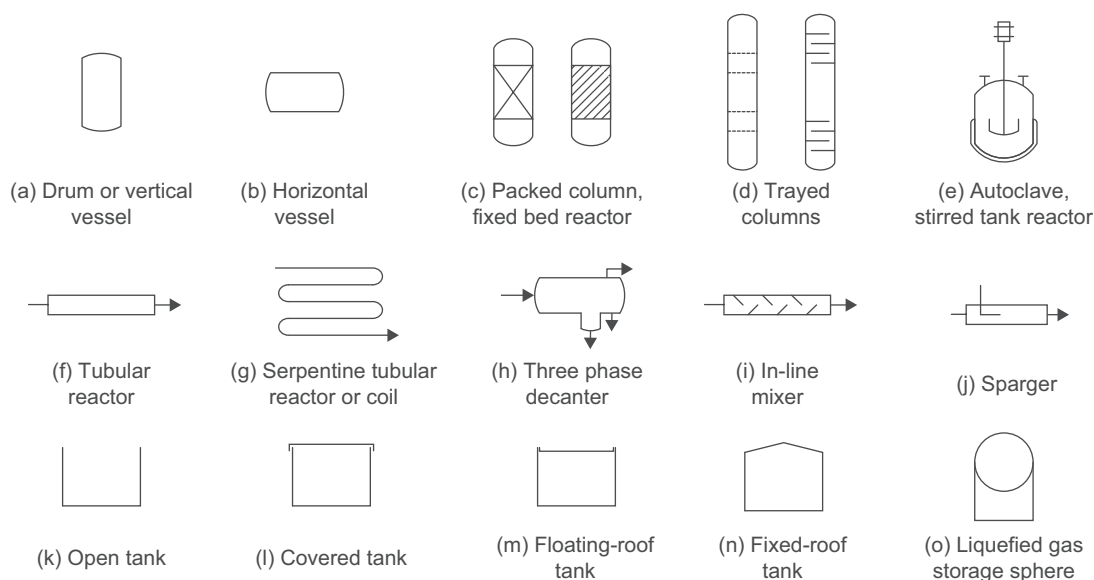


FIG. 2.2 PFD symbols for reactors, vessels, mixers, and tanks

2.2.2 PFD symbols

On the detailed flowsheets used for design and operation, the equipment is normally drawn in a stylized pictorial form. For tender documents or company brochures, actual scale drawings of the equipment are sometimes used, but it is more usual to use a simplified representation. There are several international standards for PFD symbols, but most companies use their own standard symbols, as the cost of converting all of their existing drawings would be excessive. ISO 10628 is the international standard for PFD drawing symbols. Most European countries have adopted ISO 10628 as their standard, but very few North American companies apply this standard, and there is currently no U.S. standard for PFD symbols. The symbols given in British Standard, BS 1553 (1977) "Graphical Symbols for General Engineering" Part 1, "Piping Systems and Plant" are more typical of those in common use. The professional edition of Microsoft Visio contains a library of PFD icons that includes the ISO 10628 symbols, as well as symbols commonly used in the United States and Canada. Examples of standard symbols are given in Appendix A, which is available in the online material at www.elsevier.com/books-and-journals/book-companion/9780128211793.

Fig. 2.2 shows symbols that are used for reactors, mixers, vessels, and tanks. Fig. 2.3 shows symbols used for heat transfer equipment. Fig. 2.4 provides symbols for fluid-handling equipment, and Fig. 2.5 gives symbols for solids-handling operations. Some general symbols that are used in combination with other symbols are shown in Fig. 2.6.

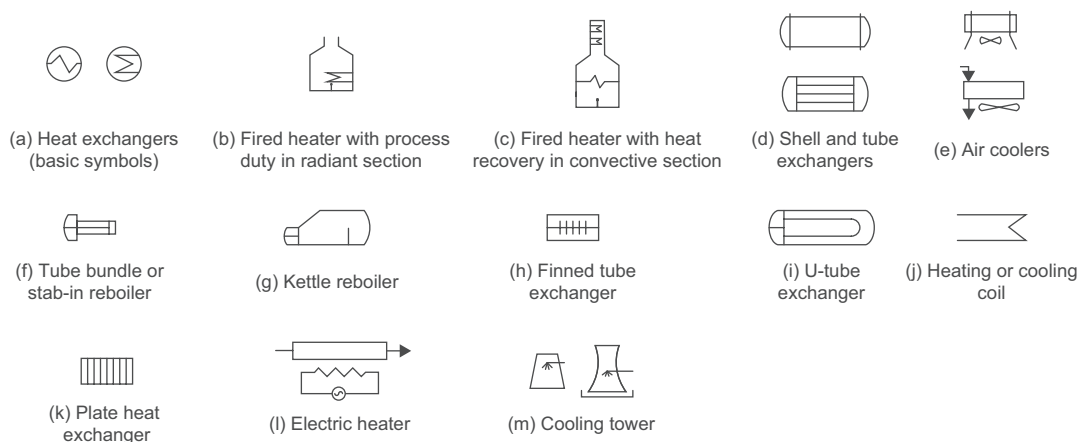


FIG. 2.3 PFD symbols for heat transfer equipment

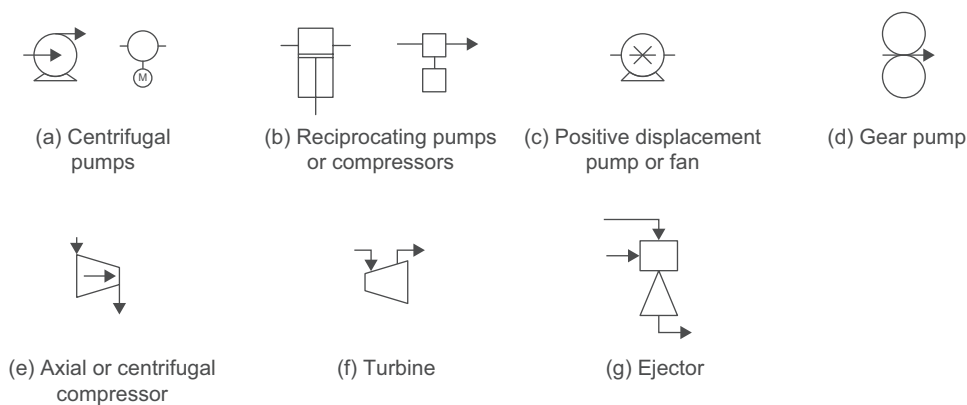


FIG. 2.4 PFD symbols for fluid-handling equipment

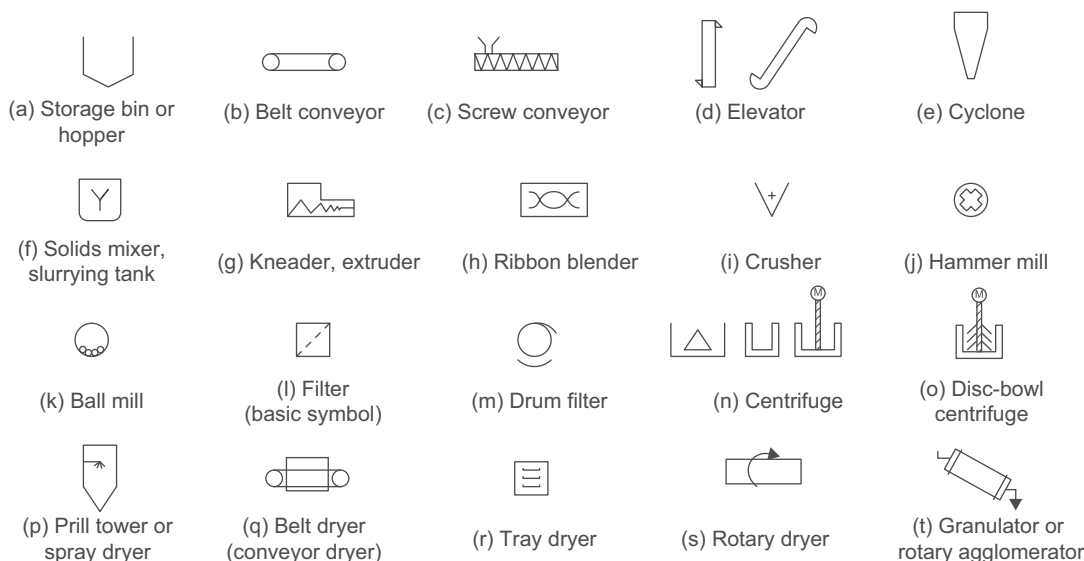


FIG. 2.5 PFD symbols for solids-handling equipment

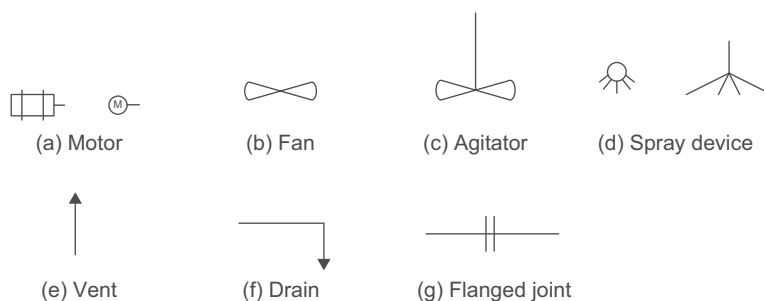


FIG. 2.6 General PFD symbols used with other symbols

The symbols that are used for process instruments, valves, and controllers are given in the section on P&ID diagram symbols in [Chapter 5](#). The operation and design of the different types of equipment illustrated in these figures are described in Part II of this book.

Note that some types of equipment have generic symbols as well as symbols that describe a particular equipment type. If the wrong symbol is selected, this can cause confusion for other engineers who read the flowsheet. For example, [Fig. 2.2\(i\)](#) shows an in-line mixer, which would be used downstream of a T-junction to ensure rapid mixing

of two liquid streams. Fig. 2.5(f) shows a solids mixer or blender that would be used to mix solids into a liquid. Fig. 2.6(c) is the symbol for a propeller agitator that might be used in a mixing tank. All of these symbols could be referred to as a *mixer*, but the designer's intention is obviously different in each case. Some abbreviations that are commonly used in PFDs are listed in the nomenclature at the end of this chapter.

2.2.3 Presentation of stream flow rates

The data on the flow rate of each individual component, on the total stream flow rate, and the percentage composition can be shown on the flowsheet in various ways. The simplest method, suitable for simple processes with few pieces of equipment, is to tabulate the data in blocks alongside the process stream lines, as shown in Fig. 2.7. Only a limited amount of information can be shown this way, and it is difficult to make neat alterations or to add more data.

A better method for the presentation of data on flowsheets is shown in Figs. 2.8 and 2.9. In this method each stream line is numbered and the data are tabulated at the bottom of the sheet. Alterations and additions can be easily made. This is the method generally used by professional design offices. A typical commercial flowsheet is shown in Fig. 2.10. Guide rules for the layout of this type of flowsheet presentation are given in Section 2.2.5.

2.2.4 Information to be included

The amount of information shown on a flowsheet will depend on the custom and practice of the particular design office. The list given here has therefore been divided into essential items and optional items. The essential items must always be shown; the optional items add to the usefulness of the flowsheet but are not always included.

Essential information

1. Always show *all* process equipment, including feed and product storage and equipment used for transporting fluids and solids.
2. Always indicate the location of process control valves.
3. For stream composition, either:
 - i. Tabulate the flow rate of each individual component, kg/h, which is preferred,
 - ii. Or give the stream composition as a weight fraction.
4. Total stream flow rate, kg/h.
5. Stream temperature, degrees Celsius preferred.
6. Nominal operating pressure (the required operating pressure).

Optional information

1. Molar percentage composition and/or molar flow rates.
2. Stream enthalpy, kJ/h.
3. Physical property data, with mean values for the stream, such as:
 - i. Density, kg/m³
 - ii. Viscosity, mN s/m²
4. Stream name: a brief one- or two-word description of the nature of the stream, for example, "ACETONE COLUMN BOTTOMS"

The convention of showing control valves on a PFD is not followed by all companies, but is recommended as a best practice. The presence of a control valve always causes pressure drop and may create a requirement for additional pumps or compressors. Showing the control valves can help identify locations where the designer may have overlooked the need for a pump.

Effective cooperation depends on effective communications, and all design organizations have formal procedures for handling project information and documentation. The project documentation will include:

1. General correspondence within the design group and with:
 - i. government departments
 - ii. equipment vendors
 - iii. site personnel
 - iv. the client

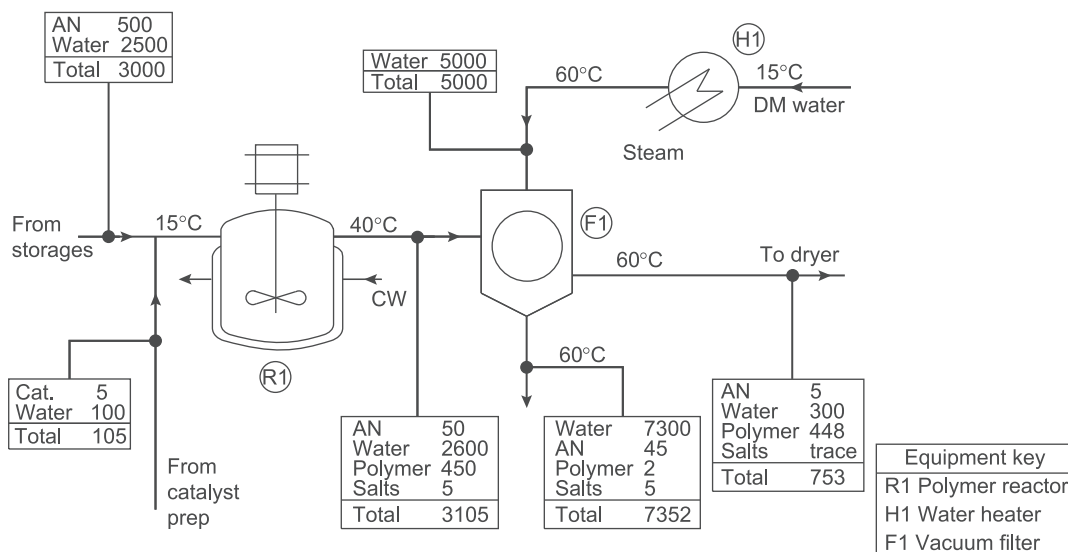


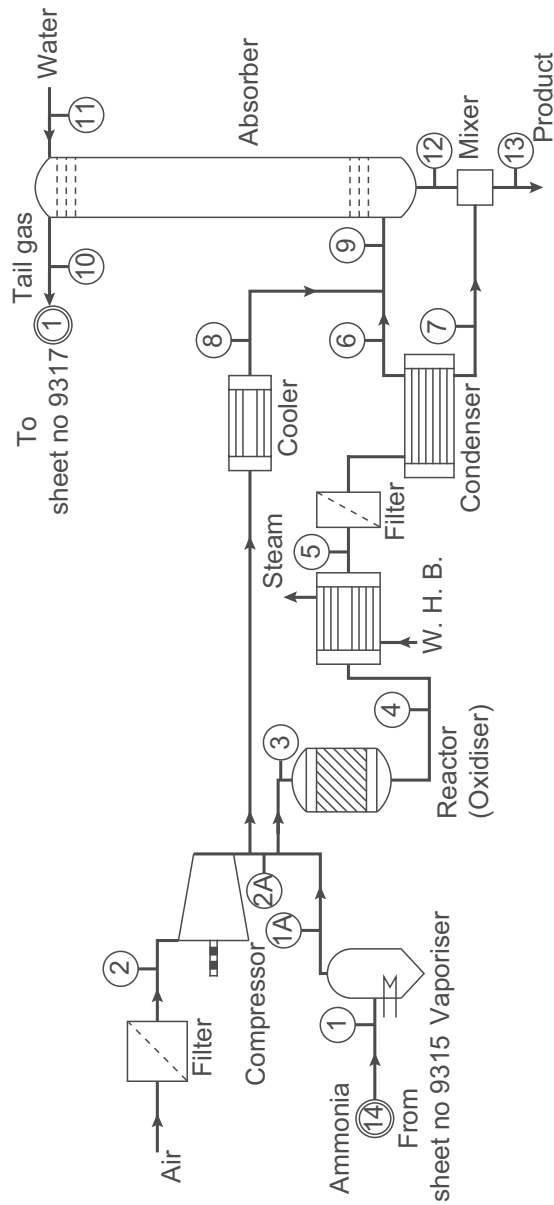
FIG. 2.7 Flowsheet: Polymer production.

2. Calculation sheets:
 - i. design calculations
 - ii. cost estimates
 - iii. material and energy balances
3. Drawings:
 - i. flow sheets
 - ii. piping and instrumentation diagrams
 - iii. layout diagrams
 - iv. plot/site plans
 - v. equipment details
 - vi. piping diagrams (isometrics)
 - vii. architectural drawings
 - viii. design sketches
4. Specification sheets:
 - i. the design basis
 - ii. feed and product specifications
 - iii. an equipment list
 - iv. sheets for equipment, such as heat exchangers, pumps, heaters, etc.
5. Health, safety, and environmental information:
 - i. materials safety data sheets (MSDS forms)
 - ii. HAZOP or HAZAN documentation (see [Chapter 10](#))
 - iii. emissions assessments and permits
5. Purchase orders:
 - i. quotations
 - ii. invoices

All documents are assigned a code number for easy cross-referencing, filing, and retrieval.

2.2.5 Layout

The sequence of the main equipment items shown symbolically on the flowsheet follows that of the proposed plant layout. Some license must be exercised in the placing of ancillary items, such as heat exchangers and pumps, or the layout will be too congested. The aim should be to show the flow of material from stage to stage as it will occur and to give a general impression of the layout of the actual process plant.



Flows kg/h pressures nominal

Line no.	Stream component	1	1A	2	2A	3	4	5	6	7	8	9	10	11	12	13	C & R Construction Inc
NH ₃	Ammonia feed	731.0	731.0	—	—	731.0	Nil	—	—	—	—	—	—	—	—	—	Nitric acid 60 percent
O ₂	—	—	—	3036.9	2628.2	2628.2	935.7	(935.7) ⁽¹⁾	275.2	Trace	408.7	683.9	371.5	—	Trace	Trace	100,000 t/y
N ₂	—	—	—	9990.8	8644.7	8644.7	8668.8	8668.8	8668.8	Trace	1346.1	10,014.7	10,014.7	—	Trace	Trace	Client BOP chemicals
NO	—	—	—	—	—	—	1238.4	(1238.4) ⁽¹⁾	202.5	—	—	202.5	21.9	—	Trace	Trace	SLIGO
NO ₂	—	—	—	—	—	—	Trace	(?) ⁽¹⁾	967.2	—	—	967.2	(Trace) ⁽¹⁾	—	Trace	Trace	Sheet no. 9316
HNO ₃	—	—	—	—	—	—	Nil	Nil	—	850.6	—	—	—	—	Trace	Trace	
H ₂ O	—	—	—	Trace	—	—	1161.0	1161.0	29.4	1010.1	—	29.4	26.3	1376.9	1704.0	2554.6	
Total		731.0	731.0	13,027.7	11,272.9	12,003.9	12,003.9	12,003.9	10,143.1	1860.7	1754.8	11,897.7	10,434.4	1376.9	2840.0	4700.6	
Press bar		8	8	1	8	8	8	8	8	1	8	8	1	8	1	1	Dwg by
Temp. °C		15	20	15	230	204	907	234	40	40	40	40	25	25	40	43	Date
																	Checked 25/7/1980

FIG. 2.8 Flowsheet: Simplified nitric acid process.



The equipment should be drawn approximately to scale. Again, some license is allowed for the sake of clarity, but the principal equipment items, such as reactors, vessels, and columns, should be drawn roughly in the correct proportion. Ancillary items can be drawn out of proportion. For a complex process with many process units, several sheets may be needed, and the continuation of the process streams from one sheet to another must be clearly shown. One method of indicating a line continuation is shown in Fig. 2.8; those lines that are continued over to another drawing are indicated by a double concentric circle around the line number, and the continuation sheet number is written below. An alternative method is to extend lines to the side of the page and then indicate the drawing sheet on which the line is continued.

The overall process flow will usually be from left to right across the diagram, with recycle streams flowing right to left. Streams should be shown entering and leaving vessels in a manner that is consistent with the stream phase, density, and mode of operation. When a vessel has multiple feeds or products, their arrangement in the diagram should correspond to the physical arrangement intended by the designer. For example, the gas feed to an absorption column should enter below the bottom tray, the liquid feed should enter on the top tray, gas product should be shown leaving the top of the vessel, and liquid product should be shown leaving the bottom of the vessel.

The equipment should be well spaced out so that streams can be labeled without the drawing becoming cluttered. It is better to use several continuation sheets than to try to fit everything on one page.

The table of stream flows and other data can be placed above or below the equipment layout. Normal practice is to place it below. The components should be listed down the left side of the table, as in Figs. 2.8 and 2.9. For a long table, it is good practice to repeat the list at the right side so the components can be traced across from either side.

The stream line numbers should follow consecutively from left to right of the layout as far as is practicable, so that when reading the flowsheet it is easy to locate a particular line and the associated column containing the data.

All the process stream lines shown on the flowsheet should be numbered and the data for the stream given. On a large flowsheet, the designers sometimes use different series of numbers for different plant sections; for example, beginning the stream numbering at 100 for feed preparation, 200 for reaction, 300 for separation, and 400 for purification. This can be helpful in quickly tracing a stream to a section of the plant. There is always a temptation to leave out the data on a process stream if it is clearly just formed by the addition of two other streams, as at a junction, or if the composition is unchanged when flowing through a process unit, such as a heat exchanger; this temptation should be avoided. What may be clear to the process designer is not necessarily clear to the others who will use the flowsheet. Complete, unambiguous information on all streams should be given, even if this involves some repetition. The purpose of the flowsheet is to show the function of each process unit, even when the function has no discernible impact on the mass and energy balance.

2.2.6 Precision of data

The total stream and individual component flows do not normally need to be shown to a high precision on the process flowsheet; three or four significant figures are all that is usually justified by the accuracy of the flowsheet calculations and will typically be sufficient. The flows should, however, balance to within the precision shown. If a stream or component flow is so small that it is less than the precision used for the larger flows, it can be shown to a greater number of places if its accuracy justifies this and the information is required. If the composition of a component is very low but is specified as a process constraint, as, say, for an effluent stream or product quality specification, it can be shown in parts per million (ppm). Imprecise small flows are best shown as "TRACE."

A trace quantity should not be shown as zero, nor should the space in the tabulation be left blank, unless the process designer *is sure* that it has no significance. The process designer should be aware that if the space in the data table is left blank opposite a particular component, the quantity may be assumed to be zero by the specialist design groups, who take their information from the flowsheet. Trace quantities can be important. Only a trace of an impurity is needed to poison a catalyst, and trace quantities can determine the selection of the materials of construction; see Chapter 6.

2.2.7 Basis of the calculation

It is good practice to show on the flowsheet the basis used for the flowsheet calculations. This includes the operating hours per year, the reaction and physical yields, and the datum temperature used for energy balances. It is also helpful to include a list of the principal assumptions used in the calculations. This alerts the user to any limitations that may have to be placed on the flowsheet information.

If the amount of information that needs to be presented is excessive, it can be summarized in a separate document that is referenced on the flowsheet.

In some cases, mass and energy balances are prepared for multiple scenarios. These might include winter and summer operating conditions, start of catalyst life and end of catalyst life, manufacture of different products or product grades, etc. Usually these different scenarios are shown as several tables on the same flowsheet, but occasionally different flowsheets are drawn for each case.

2.2.8 Batch processes

Flowsheets drawn up for batch processes normally show the quantities required to produce one batch. If a batch process forms part of an otherwise continuous process, it can be shown on the same flowsheet, providing a clear break is made when tabulating the data between the continuous and batch sections (i.e., the change from kg/h to kg/batch).

A continuous process may include batch make-up of minor reagents, such as the catalyst for a polymerization process. Batch flows into a continuous process are usually labeled “Normally no flow” and show the flow rates that will be obtained when the stream is flowing. It is these instantaneous flow rates that govern the equipment design rather than the much lower time-averaged flow rates.

2.2.9 Utilities

To avoid cluttering up the flowsheet, it is not normal practice to show the utility (service) headers and lines on the process flowsheet. The utility connections required on each piece of equipment should be shown and labeled, for example, “CTW” for cooling tower water. The utility requirements for each piece of equipment should be tabulated on the flowsheet. A glossary of common abbreviations used for utility streams in PFDs is given in the nomenclature at the end of this chapter. Utility systems are described in more detail in [Chapter 3](#).

2.2.10 Equipment identification

Each piece of equipment shown on the flowsheet must be identified with a code number and name. The identification number (usually a letter and some digits) is normally that assigned to a particular piece of equipment as part of the general project control procedures and is used to identify it in all the project documents.

If the flowsheet is not part of the documentation for a project, then a simple, but consistent, identification code should be devised. The easiest code is to use an initial letter to identify the type of equipment, followed by digits to identify the particular piece. For example, H — heat exchangers, C — columns, R — reactors. Most companies have a standard convention that should be followed, but if there is no agreed standard, then the key to the code should be shown on the flowsheet.

2.2.11 Flowsheet drafting programs

Most design offices use drafting software for the preparation of flowsheets and other process drawings. With drafting software, standard symbols representing the process equipment, instruments, and control systems are held in files, and these symbols are called up as required when drawing flowsheets and piping and instrumentation diagrams. Final flowsheet drawings are usually produced by professional drafters, who are experienced with the drafting software and conventions, rather than by the design engineer. The design engineer has to provide the required numbers, sketch the flowsheet, and review the final result.

Although most process simulation programs feature a graphical user interface (GUI) that creates a drawing that resembles a PFD, printouts of these drawings should not be used as actual PFDs. The unit operations shown in the process simulation usually do not exactly match the unit operations of the process. The simulation may include dummy items that do not physically exist and may omit some equipment that is needed in the plant but is not part of the simulation.

2.3 The anatomy of a chemical manufacturing process

This section describes the basic components of chemical processes and discusses how designers select between batch and continuous processes. The effects of reactor yield and selectivity on flowsheet structure are examined and used to illustrate how flowsheets become complex when there are multiple feeds and products.

2.3.1 Components of a chemical process

The basic components of a typical chemical process are shown in Fig. 2.11, in which each block represents a stage in the overall process for producing a product from the raw materials. Fig. 2.11 represents a generalized process; not all the stages will be needed for any particular process, and the complexity of each stage will depend on the nature of the process. Chemical engineering design is concerned with the selection and arrangement of the stages and the selection, specification, and design of the equipment required to perform the function of each stage.

Stage 1. Raw material storage

Unless the raw materials (also called *feed stocks* or *feeds*) are supplied as intermediate products (intermediates) from a neighboring plant, some provision will have to be made to hold several days', or weeks', worth of storage to smooth out fluctuations and interruptions in supply. Even when the materials come from an adjacent plant, some provision is usually made to hold inventory for a few hours, or even days, to decouple the processes. The storage required depends on the nature of the raw materials, the method of delivery, and what assurance can be placed on the continuity of supply. If materials are delivered by ship (tanker or bulk carrier) several weeks' worth of stocks may be necessary, whereas if they are received by road or rail in smaller lots, less storage will be needed.

Stage 2. Feed preparation

Some purification and preparation of the raw materials will usually be necessary before they are sufficiently pure, or in the right form, to be fed to the reaction stage. For example, acetylene generated by the carbide process contains arsenic and sulfur compounds and other impurities, which must be removed by scrubbing with concentrated sulfuric acid (or other processes) before it is sufficiently pure for reaction with hydrochloric acid to produce dichloroethane. Feed contaminants that can poison process catalysts, enzymes, or microorganisms must be removed. Liquid feeds need to be vaporized before being fed to gas-phase reactors, and solids may need crushing, grinding, and screening. Solid feeds may also need to be weighed and mixed into slurries or solutions so that they can be brought to process pressure and easily mixed with other components.

The feed preparation stage always includes means for getting the feeds out of storage and into the process. Liquids are usually pumped out of storage through control valves that regulate the feed flow rate. Gases and vapors may need compression if the storage is not pressurized. Solids are conveyed from storage using a variety of equipment, described in Chapter 18.

Biological processes require careful feed preparation. The growth media and any other fluids that are fed to the cell culture must be sterile to prevent unwanted organisms from entering the process. Sterilization is usually accomplished by heating the feed to a high temperature and keeping it hot for long enough to kill unwanted organisms, then cooling the feed to the desired temperature for the reactor. The preparation of biological reactor feeds is discussed in more detail in Section 15.9.

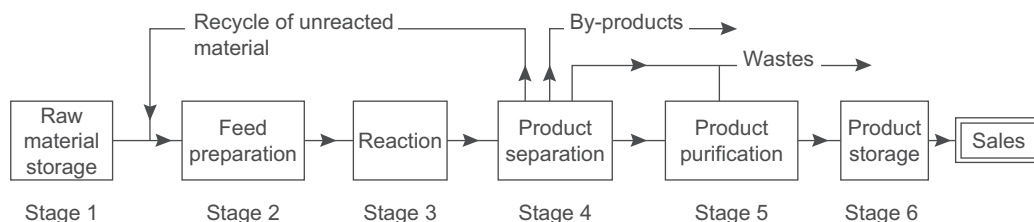


FIG. 2.11 Anatomy of a chemical process.

Stage 3. Reaction

The reaction stage is the heart of a chemical manufacturing process. In the reactor the raw materials are brought together under conditions that promote the production of the desired product; almost invariably, some by-products will also be formed, either through the reaction stoichiometry, by side reactions, or from reactions of impurities present in the feed. Reactor design is discussed in [Chapter 15](#).

Stage 4. Product separation

After the reactor(s), the products and by-products are separated from any unreacted material. If in sufficient quantity, the unreacted material will be recycled to the reaction stage or to the feed purification and preparation stage. The by-products may also be separated from the products at this stage and may undergo further processing for recovery or sale. In most chemical processes there are multiple reaction steps, each followed by one or more separation steps.

Stage 5. Purification

Before sale, the main product will often need purification to meet the product specifications. If produced in economic quantities, the by-products may also be purified for sale. For by-products, there will always be an economic trade-off between purifying the by-product for sale or disposing of it as recycle or waste.

Stage 6. Product storage

Some inventory of finished product must be held to match production with sales. Provision for product packaging and transport is also needed, depending on the nature of the product. Liquids are normally dispatched in drums and in bulk tankers (road, rail, and sea); solids in sacks, cartons, or bales.

The amount of stock that is held will depend on the nature of the product and the market.

Ancillary processes

In addition to the main process stages shown in [Fig. 2.11](#), provision must be made for the supply of the utilities needed, such as process water, cooling water, compressed air, and steam. The design of utility systems is discussed in [Chapter 3](#).

2.3.2 Continuous and batch processes

Continuous processes are designed to operate 24 hours a day, 7 days a week, throughout the year. Some downtime will be allowed for maintenance and, in some processes, for catalyst regeneration. The plant attainment or operating rate is the percentage of the available hours in a year that the plant operates and is usually between 90% and 95%.

$$\text{Attainment \%} = \frac{\text{hours operated}}{8760} \times 100 \quad (2.1)$$

A typical design basis would assume 8000 operating hours per year.

Batch processes are designed to operate intermittently, with some, or all, of the process units being frequently shut down and started up. It is quite common for batch plants to use a combination of batch and continuous operations. For example, a batch reactor may be used to feed a continuous distillation column.

Continuous processes will usually be more economical for large-scale production. Batch processes are used when some flexibility is wanted in production rate or product specifications.

The advantages of batch processing are:

- Batch processing allows production of multiple different products or different product grades in the same equipment.
- In a batch plant, the integrity of a batch can be preserved as it moves from operation to operation. This can be useful for quality control purposes.
- The production rate of batch plants is flexible, as there are no turn-down issues when operating at low output.
- Batch plants are easier to clean and maintain sterile operation.
- Batch processes are easier to scale up from chemist's recipes.
- Batch plants have low capital for small production volumes. The same piece of equipment can often be used for several unit operations.

The drawbacks of batch processing are:

- The scale of production is limited.
- It is difficult to achieve economies of scale by going to high production rates.
- Batch-to-batch quality can vary, leading to high production of waste products or off-spec product.
- Recycle and heat recovery are harder, making batch plants less energy efficient and more likely to produce waste by-products.
- Asset utilization is lower for batch plants, as the plant almost inevitably is idle part of the time.
- Batch plants are more labor-intensive, and so the fixed costs of production are much higher for batch plants on a \$/unit mass of product basis.

Choice of continuous versus batch production

Given the higher fixed costs and lower plant utilization of batch processes, batch processing usually only makes sense for products that have high value and are produced in small quantities. Batch plants are commonly used for:

- Food products
- Pharmaceutical products such as drugs, vaccines, and hormones
- Personal care products
- Blended products with multiple grades, such as paints, detergents, etc.
- Specialty chemicals

Even in these sectors, continuous production is favored if the process is well understood, the production volume is large, and the market is competitive.

2.3.3 Effect of reactor conversion and yield on flowsheet structure

It is important to distinguish between conversion and yield. Conversion has to do with reactants; yield with products.

Conversion

Conversion is a measure of the fraction of the reagent that reacts. To optimize reactor design and minimize by-product formation, the conversion of a particular reagent is often less than 100%. If more than one reactant is used, the reagent on which the conversion is based must be specified.

Conversion is defined by the following expression:

$$\begin{aligned}\text{Conversion} &= \frac{\text{amount of reagent consumed}}{\text{amount supplied}} \\ &= \frac{(\text{amount in feed stream}) - (\text{amount in product stream})}{(\text{amount in feed stream})}\end{aligned}\tag{2.2}$$

This definition gives the total conversion of the particular reagent to all products.

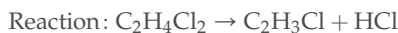
Example 2.1

In the manufacture of vinyl chloride (VC) by the pyrolysis of dichloroethane (DCE), the reactor conversion is limited to 55% to reduce carbon formation, which fouls the reactor tubes.

Calculate the quantity of DCE fed to the reactor to produce 5000 kg/h VC.

Solution

Basis: 5000 kg/h VC (the required quantity).



Molar weights: DCE 99, VC 62.5

$$\text{kmol/h VC produced} = \frac{5000}{62.5} = \underline{\underline{80}}$$

From the stoichiometric equation, 1 kmol DCE produces 1 kmol VC. Let X be DCE feed in kmol/h:

$$\text{Percent conversion} = 55 = \frac{80}{X} \times 100$$

$$X = \frac{80}{0.55} = \underline{\underline{145.5 \text{ kmol/h}}}$$

In this example, the small loss of DCE to carbon and other products has been neglected. All the DCE reacted has been assumed to be converted to VC.

Selectivity

Selectivity is a measure of the efficiency of the reactor in converting reagent to the desired product. It is the fraction of the reacted material that was converted into the desired product. If no by-products are formed, then the selectivity is 100%. If side reactions occur and by-products are formed, then the selectivity decreases. Selectivity is always expressed as the selectivity of feed A for product B and is defined by the equation:

$$\begin{aligned} \text{Selectivity} &= \frac{\text{moles of B formed}}{\text{moles of B that could have been formed if all A reacted to give B}} \\ &= \frac{\text{moles of B formed}}{\text{moles of A consumed} \times \text{stoichiometric factor}} \end{aligned} \quad (2.3)$$

Stoichiometric factor = moles of B produced per mole of A reacted in the reaction stoichiometric equation

Selectivity is usually improved by operating the reactor at low conversion. At high conversion, the reactor has low concentrations of at least one reagent and high concentrations of products, so reactions that form by-products are more likely to occur.

Reagents that are not converted in the reactor can be recovered and recycled. Reagents that become converted to by-products usually cannot be recovered, and the by-products must be purified for sale or else disposed as waste (see Section 8.2.3). The optimum reactor conditions thus usually favor low reactor conversion to give high selectivity for the desired products when all of these costs are taken into account.

Yield

Yield is a measure of the performance of a reactor or plant. Several different definitions of yield are used, and it is important to clearly state the basis of any yield numbers. This is often not done when yields are quoted in the literature, and judgment must be used to decide what was intended.

The yield of product B from feed A is defined by:

$$\text{Yield} = \frac{\text{moles of B formed}}{\text{moles of A supplied} \times \text{stoichiometric factor}} \quad (2.4)$$

For a reactor, the yield is the product of conversion and selectivity:

$$\begin{aligned} \text{Reaction yield} &= \text{Conversion} \times \text{Selectivity} \\ &= \frac{\text{moles A consumed}}{\text{moles A supplied}} \times \frac{\text{moles B formed}}{\text{moles A consumed} \times \text{stoichiometric factor}} \end{aligned} \quad (2.5)$$

With industrial reactors, it is necessary to distinguish between “reaction yield” (chemical yield), which includes only chemical losses to side products, and the overall “reactor yield,” which also includes physical losses, such as losses by evaporation into vent gas.

If the conversion is near 100% it may not be worth separating and recycling the unreacted material; the overall process yield would then include the loss of unreacted material. If the unreacted material is separated and recycled, the overall process yield *taken over the reactor and separation step* would include any physical losses from the separation step.

Plant yield is a measure of the overall performance of the plant and includes all chemical and physical losses.

Plant yield (applied to the complete plant or any stage):

$$= \frac{\text{moles of product produced}}{\text{moles of reagent supplied to the process} \times \text{stoichiometric factor}} \quad (2.6)$$

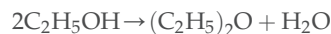
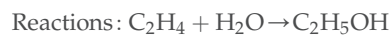
Where more than one reagent is used or product produced, it is essential that the product and reagent to which the yield refers are clearly stated.

The plant yield of B from A is the product of the reactor selectivity of feed A for product B and the separation efficiency (recovery) of each separation step that handles product B or reagent A. As a useful check, the plant yield should be greater than the reactor yield if a separation and feed recycle scheme has been implemented. If the feed recovery and recycle system were 100% efficient, the plant yield would approach the reactor selectivity.

Example 2.2

In the production of ethanol by the hydrolysis of ethylene, diethyl ether is produced as a by-product. A typical feed stream composition is 55% ethylene, 5% inerts, 40% water, and product stream: 52.26% ethylene, 5.49% ethanol, 0.16% ether, 36.81% water, and 5.28% inerts. Calculate the selectivity of ethylene for ethanol and for ether.

Solution



Basis: 100 moles feed (easier calculation than using the product stream)

Note: The flow of inerts will be constant, as they do not react, and it can thus be used to calculate the other flows from the compositions.

Feed stream

ethylene	55 mol
inerts	5 mol
water	40 mol

Product stream

$$\text{ethylene} = \frac{52.26}{5.28} \times 5 = 49.49 \text{ mol}$$

$$\text{ethanol} = \frac{5.49}{5.28} \times 5 = 5.20 \text{ mol}$$

$$\text{ether} = \frac{0.16}{5.28} \times 5 = 0.15 \text{ mol}$$

$$\text{Amount of ethylene reacted} : = 55.0 - 49.49 = 5.51 \text{ mol}$$

$$\text{Selectivity of ethylene for ethanol} : = \frac{5.20}{5.51 \times 1.0} \times 100 = \underline{\underline{94.4\%}}$$

As 1 mol of ethanol is produced per mol of ethylene, the stoichiometric factor is 1.

$$\text{Selectivity of ethylene for ether} : = \frac{0.15}{5.51 \times 0.5} \times 100 = \underline{\underline{5.44\%}}$$

The stoichiometric factor is 0.5, as 2 mol of ethylene produce 1 mol of ether.

Note that the conversion of ethylene to all products is given by:

$$\begin{aligned} \text{Conversion} &= \frac{\text{mols fed} - \text{mols out}}{\text{mols fed}} = \frac{55 - 49.49}{55} \times 100 \\ &= \underline{\underline{10 \text{ percent}}} \end{aligned}$$

The selectivity based on water could also be calculated but is of no real interest, as water is relatively inexpensive compared with ethylene. Water is clearly fed to the reactor in considerable excess.

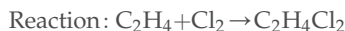
The yield of ethanol based on ethylene is:

$$\text{Reaction yield} = \frac{5.20}{55 \times 1.0} \times 100 = \underline{\underline{9.45\%}}$$

Example 2.3

In the chlorination of ethylene to produce DCE, the conversion of ethylene is reported as 99.0%. If 94 mol of DCE are produced per 100 mol of ethylene reacted, calculate the selectivity and the overall yield based on ethylene. The unreacted ethylene is not recovered.

Solution



The stoichiometric factor is 1.

$$\begin{aligned} \text{Selectivity} &= \frac{\text{moles DCE produced}}{\text{moles ethylene reacted} \times 1} \times 100 \\ &= \frac{94}{100} \times 100 = \underline{\underline{94\%}} \end{aligned}$$

$$\text{Overall yield (including physical losses)} = \frac{\text{moles DCE produced}}{\text{moles ethylene fed} \times 1} \times 100$$

Ninety-nine moles of ethylene are reacted for 100 moles fed, so

$$\text{Overall yield} = \frac{94}{100} \times \frac{99}{100} = \underline{\underline{93.1\%}}$$

Note that we get the same answer by multiplying the selectivity (0.94) and conversion (0.99).

The principal by-product of this process is trichloroethane.

Effect of conversion, selectivity, and yield on flowsheet structure

Very few processes produce the desired product in stoichiometric yield with no by-products and no equilibrium limitations.

If the desired reaction is limited by equilibrium between the feeds and products, the reaction will not proceed to 100% conversion in the reactor, and it will be necessary to separate the product from unreacted feed components. It will usually be economically attractive to recover the unconverted feeds either to the reactor section or to the feed preparation section of the plant.

In most chemical processes the designer must also address the formation of by-products through unwanted reactions, in which case the selectivity of the feed for the desired product is less than 100%. The presence of nonselective reactions can have a number of undesirable effects on process economics. The most important impact of by-product formation is that by-products represent a loss of potential product. Because feedstock costs are usually the main component of the overall cost of production, low selectivity can have a strong negative impact on process economics. The by-products must be separated from the desired product, causing additional complexity and cost in the separation section. If the by-products have value, they can be purified and sold, but this adds more equipment to the process. If the by-products are not worth recovering, then in some cases they can be recycled within the process and converted back to feed or products. These recycles also add cost and complexity to the process. If the by-products cannot be sold or recycled, they must be disposed of as waste streams. Additional processing steps may be needed to bring the waste stream to a safe condition for discharge or disposal.

Because of the high costs of dealing with by-products, most processes are operated under conditions that maximize reactor selectivity. This often means operating at low conversion and accepting large recycles of feeds. Alternatively, a cheaper feed may be used in excess, so that a high conversion of the more expensive feed can be achieved at high selectivity, as discussed in the following section.

Use of excess reagent

In industrial reactions the components are seldom fed to the reactor in exact stoichiometric proportions. A reagent may be supplied in excess to promote the desired reaction; to maximize the use of an expensive reagent; or to ensure complete reaction of a reagent, as in combustion.

The percentage excess reagent is defined by the following equation:

$$\text{Percent excess} = \frac{\text{quantity supplied} - \text{stoichiometric}}{\text{stoichiometric quantity}} \times 100 \quad (2.7)$$

It is necessary to state clearly to which reagent the excess refers.

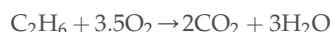
Example 2.4

To ensure complete combustion, 20% excess air is supplied to a furnace burning natural gas. The gas composition (by volume) is methane 95% and ethane 5%.

Calculate the moles of air required per mole of fuel.

Solution

Basis: 100 mol gas, as the analysis is volume percentage.



$$\text{Stoichiometric moles of O}_2 \text{ required} = 95 \times 2 + 5 \times 3.5 = \underline{207.5}$$

$$\text{With 20\% excess, moles of O}_2 \text{ required} = 207.5 \times 1.2 = 249$$

$$\text{Moles of air (21\% O}_2) = 249 \times 100/21 = 1185.7$$

$$\text{Air per mole of fuel} = 1185.7 / 100 = \underline{11.86 \text{ mol}}$$

Sources of conversion, selectivity, and yield data

If there is minimal by-product formation, then the reactor costs (volume, catalyst, heating, etc.) can be traded off against the costs of separating and recycling unconverted reagents to determine the optimal reactor conversion. More frequently, the selectivity of the most expensive feeds for the desired product is less than 100%, and by-product costs must also be taken into account. The reactor optimization then requires a relationship between reactor conversion and selectivity, not just for the main product but for all the by-products that are formed in sufficient quantity to have an impact on process costs.

In simple cases, when the number of by-products is small, it may be possible to develop a mechanistic model of the reaction kinetics that predicts the rate of formation of the main product and by-products. If such a model is fitted to experimental data over a suitably wide range of process conditions, then it can be used for process optimization. The development of reaction kinetics models is discussed in Section 15.3 and is described in most reaction engineering textbooks. See, for example, [Levenspiel \(1998\)](#), [Froment et al. \(2010\)](#), [Fogler \(2016\)](#), and [Rawlings and Eckerdt \(2013\)](#).

In cases where the reaction quickly proceeds to equilibrium, the yields are easily estimated as the equilibrium yields. Under these circumstances, the only possibilities for process optimization are to change the temperature, pressure, or feed composition so as to obtain a different equilibrium mixture. The calculation of reaction equilibrium is easily carried out using commercial process simulation programs, as described in Section 4.5.1.

When the number of components or reactions is too large, or the mechanism is too complex to deduce with statistical certainty, then response surface models can be used instead. Methods for the statistical design of experiments can be applied, reducing the amount of experimental data that must be collected to form a statistically meaningful correlation of selectivity and yield to the main process parameters. See [Montgomery \(2019\)](#) for a good introduction to the statistical design of experiments.

In the early stages of design, the design engineer will often have neither a response surface nor a detailed mechanistic model of the reaction kinetics. Few companies are prepared to dedicate a laboratory or pilot plant and the necessary staff to collecting reaction kinetics data until management has been satisfied that the process under investigation is economically attractive. A design is thus needed before the necessary selectivity and yield data set has been collected. Under such circumstances, the design engineer must select the optimal reactor conditions from whatever data are available. This initial estimate of reactor yield may come from a few data points collected by a chemist or taken from a patent or research paper. The use of data from patents is discussed in Section 2.4.1.

For the purposes of completing a design, only a single estimate of reactor yield is needed. Additional yield data taken over a broader range of process conditions give the designer greater ability to properly optimize the design. In process synthesis projects, one purpose of the design may be to set yield targets for a research team, as described in Section 2.6.1.

2.3.4 Recycles and purges

Processes in which a flow stream is returned (recycled) to an earlier stage in the processing sequence are common. If the conversion of a valuable reagent in a reaction process is appreciably less than 100%, the unreacted material is usually separated and recycled.

Separation processes can also be a source of recycles. The return of reflux to the top of a distillation column is an example of a recycle process in which there is no reaction.

The presence of recycle streams makes the calculation of process material and energy balances more difficult. Without recycle, the material balances on a series of processing steps that can be carried out sequentially, taking each unit in turn, with the calculated flows out of one unit becoming the feeds to the next. If a recycle stream is present, then at the point where the recycle is returned, the flow will not be known, as it will depend on downstream flows not yet calculated. Without knowing the recycle flow, the sequence of calculations cannot be continued to the point where the recycle flow can be determined.

Two approaches to the solution of recycle problems are possible:

1. **The cut and try ("tear") method.** The recycle stream flows can be estimated and the calculations continued to the point where the recycle is calculated. The estimated flows are then compared with those calculated, and a better estimate is made. The procedure is continued until the difference between the estimated and the calculated flows is within an acceptable tolerance.
2. **The formal algebraic method.** The presence of recycle implies that some of the mass balance equations must be solved simultaneously. The equations are set up with the recycle flows as unknowns and solved using standard methods for the solution of simultaneous equations.

With simple problems that have only one or two recycle loops, the calculation can often be simplified by the careful selection of the basis of calculation and the system boundaries. This is illustrated in Example 2.5.

The solution of more complex material balance problems involving several recycle loops is discussed in Chapter 4.

Example 2.5

The block diagram in Fig. 2.12 shows the main steps in the balanced process for the production of vinyl chloride from ethylene. Each block represents a reactor and several other processing units. The main reactions are:

Block A, chlorination

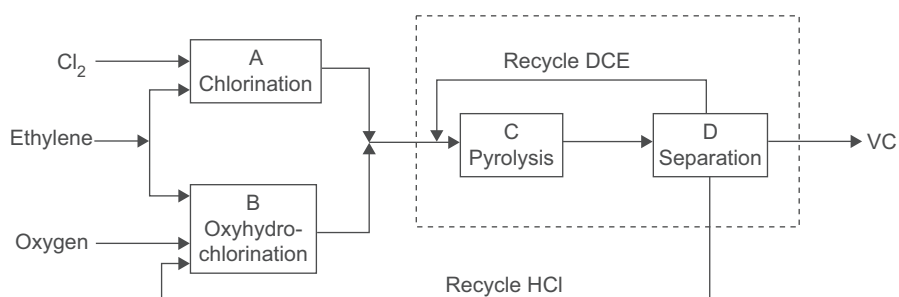


FIG. 2.12 Block flow diagram of balanced process for vinyl chloride (VC).

Block B, oxyhydrochlorination**Block C, pyrolysis**

The HCl from the pyrolysis step is recycled to the oxyhydrochlorination step. The flow of ethylene to the chlorination and oxyhydrochlorination reactors is adjusted so that the production of HCl is in balance with the requirement. The conversion in the pyrolysis reactor is limited to 55%, and the unreacted DCE is separated and recycled.

Using the yields given, and neglecting any other losses, calculate the flow of ethylene to each reactor and the flow of DCE to the pyrolysis reactor, for a production rate of 12,500 kg/h VC.

Solution

Molecular weights: vinyl chloride 62.5, DCE 99.0, HCl 36.5.

$$\text{VC per hour} = \frac{12,500}{62.5} = 200 \text{ kmol/h}$$

Draw a system boundary around each block, enclosing the separation section (block D) and the DCE recycle within the boundary of step C, as shown in Fig. 2.12.

Let the flow of ethylene to block A be X and to block B be Y and the HCl recycle be Z .

Then the total moles of DCE produced = $0.98X + 0.95Y$, allowing for the yields, and the moles of HCl produced in block C

$$= (0.98X + 0.95Y)0.995 = Z \quad (a)$$

Consider the flows to and from block B. The yield of DCE based on HCl is 90%, so the moles of DCE produced:

$$= 0.5 \times 0.90Z$$

Note: The stoichiometric factor is 0.5 (2 mol HCl per mol DCE).

The yield of DCE based on ethylene is 95%, so

$$0.5 \times 0.9Z = 0.95Y$$

$$Z = 0.95 \times 2Y/0.9$$

Substituting for Z into equation (a) gives

$$Y = (0.98X + 0.95Y)0.995 \times \frac{0.9}{2 \times 0.95} \quad (b)$$

$$Y = 0.837X$$

Turning to block C, total VC produced = $0.99 \times$ total DCE, so

$$0.99(0.98X + 0.95Y) = 200 \text{ kmol/h}$$

Substituting for Y from equation (b) gives $X = \underline{\underline{113.8 \text{ kmol/h}}}$

$$\text{and } Y = 0.837 \times 113.8 = \underline{\underline{95.3 \text{ kmol/h}}}$$

HCl recycle from equation (a)

$$Z = (0.98 \times 113.8 + 0.95 \times 95.3)0.995 = \underline{\underline{201.1 \text{ kmol/h}}}$$

$$\text{Note: Overall yield on ethylene} = \frac{200}{(113.8 + 95.3)} \times 100 = \underline{\underline{96 \text{ percent}}}$$

The total flow of DCE from blocks A and B is $\frac{200}{0.99} = 202 \text{ kmol/h}$, but this does not include the recycle. Because the conversion is 55%, if the recycle flow is R , then $202 / (202 + R) = 0.55$; hence the total flow of DCE to the pyrolysis reactor is $202 + R = 202 / 0.55 = 367.3 \text{ kmol/h}$

Purge

It is usually necessary to bleed off a portion of a recycle stream to prevent the buildup of unwanted material. For example, if a reactor feed contains inert components or by-products that are not separated from the recycle stream in the separation units, these inerts would accumulate in the recycle stream until the stream eventually consisted entirely of inerts. Some portion of the stream must be purged to keep the inert level within acceptable limits. A continuous purge would normally be used. Under steady-state conditions:

$$\text{Loss of inert in the purge} = \text{Rate of feed of inerts into the system}$$

The concentration of any component in the purge stream is the same as that in the recycle stream at the point where the purge is taken off. So the required purge rate can be determined from the following relationship:

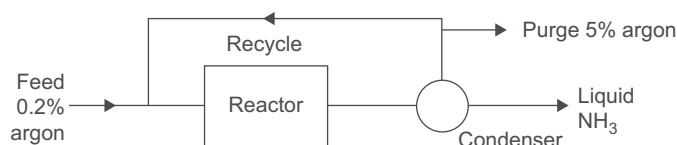
$$\begin{aligned} & [\text{Feed stream flow} - \text{rate}] \times [\text{Feed stream inert concentration}] \\ &= [\text{Purge stream flow} - \text{rate}] \times [\text{Specified(desired) recycle inert concentration}] \end{aligned}$$

Example 2.6

In the production of ammonia from hydrogen and nitrogen, the conversion, based on either raw material, is limited to 15%. The ammonia produced is condensed from the reactor (converter) product stream, and the unreacted material is recycled. If the feed contains 0.2% argon (from the nitrogen separation process), calculate the purge rate required to hold the argon in the recycle stream below 5.0%. Percentages are by volume.

Solution

Basis: 100 moles feed (purge rate will be expressed as moles per 100 mol feed, as the production rate is not given).
Process diagram:



Volume percentages are taken as equivalent to mol%.

Argon entering system with feed = $100 \times 0.2/100 = 0.2$ mol.

Let purge rate per 100 mol feed be F .

Argon leaving system in purge = $F \times 5/100 = 0.05 F$.

At the steady state, argon leaving = argon entering

$$0.05F = 0.2$$

$$F = \frac{0.2}{0.05} = \underline{4}$$

Purge required: 4 mol per 100 mol feed.

Bypass

A flow stream may be divided and some part diverted (bypassed) around some units. This procedure is often used to control stream composition or temperature.

Material balance calculations on processes with bypass streams are similar to those involving recycle, except that the stream is fed forward instead of backward. This usually makes the calculations easier than with recycle.

2.4 Selection, modification, and improvement of commercially proven processes

Engineers in industry do not usually design a new process from scratch if a commercially proven alternative is available. Companies usually seek to avoid the extra costs and risks inherent in technology commercialization. New molecules are usually made using adaptations of processes that have been shown to work for similar compounds. Even when a brand-new process is contemplated, the design team will usually also prepare a conventional design for comparison.

The use of a proven basic flow scheme does not eliminate innovation from the design. Several alternative designs may already be in commercial practice, each optimized around different feeds, catalysts, or reactor concepts. The design team must evaluate the different designs and optimize each to the local design basis to select the best. The commercial processes may need modification to make the desired product or by-products or to process an unusual feed material. It may be possible to improve the existing technology by substituting one or more unit operations, by using better catalysts or enzymes, by deploying improved separation or reactor technology, or by using different solvents to reduce environmental impact. The scale of production may also cause changes to the flowsheet; for example, if a large new plant requires reactors or separation columns to be constructed in parallel trains.

This section discusses factors a design team should consider when developing a flowsheet based on a commercially proven technology. The special case of developing a flowsheet for the revamp of an existing plant is treated in Section 2.5.

2.4.1 Sources of information on manufacturing processes

This section gives a brief overview of sources of information on commercial processes that can be found in the open literature.

The chemical process industries are competitive, and the information that is published on commercial processes is restricted. The articles on particular processes published in the technical literature and in textbooks invariably give only a superficial account of the chemistry and unit operations used. They lack the detailed information on reaction kinetics, process conditions, equipment parameters, and physical properties that is needed for process design. The information that can be found in the general literature is, however, useful in the early stages of a project, when searching for possible process routes. It is often sufficient for a flowsheet of the process to be drawn up and a rough estimate of the capital and production costs made.

The most comprehensive collection of information on manufacturing processes is probably the *Encyclopedia of Chemical Technology* edited by Kirk and Othmer (2001), which covers the whole range of chemical and associated products. An abridged version of the Kirk–Othmer encyclopedia is also available in paperback (Kirk, 2007). The latest version of the Kirk–Othmer encyclopedia is available through the Wiley online library at <http://onlinelibrary.wiley.com>. Another encyclopedia covering manufacturing processes is that edited by McKetta (2001). Several books have also been published that give brief summaries of the production processes used for the commercial chemicals and chemical products. The best known of these is probably Shreve's book on the chemical process industries, now updated by Austin and Basta (1998). Comyns (2019) lists named chemical manufacturing processes, with references.

The extensive German reference work on industrial processes, *Ullman's Encyclopedia of Industrial Technology*, is now available in an English translation, Ley, C., (Ed.) (2002).

Specialized texts have been published on some of the more important bulk industrial chemicals, such as that by Miller (1969) on ethylene and its derivatives; these are too numerous to list but should be available in the larger reference libraries and can be found by reference to the library catalogue. Meyers (2016) gives a good introduction to the processes used in oil refining. Kohl and Nielsen (1997) provide an excellent overview of the processes used for gas treating and sulfur recovery.

Many of the references cited here are available in electronic format from Knovel. Most companies and universities have Knovel subscriptions. Access to Knovel is also available to members of professional societies such as the American Institute of Chemical Engineers (AIChE).

Books quickly become outdated, and many of the processes described are obsolete, or at best obsolescent. More up-to-date descriptions of the processes in current use can be found in the technical journals. The journal *Hydrocarbon Processing* publishes an annual review of petrochemical processes, which was entitled *Petrochemical Developments* and is now called *Petrochemicals Notebook*; this gives flow diagrams and brief process descriptions of new process developments.

Patents

Patents can be a useful source of information, but some care is needed in extracting information from them. To obtain a patent, an inventor is legally obliged to disclose the best mode of practice of the invention; failure to do so could render the patent invalid if it were contested. Most patents therefore include one or more examples illustrating how the invention is practiced and differentiating it from the prior art. The examples given in a patent often give an indication of the process conditions used, though they are frequently examples of laboratory preparations rather than of the full-scale manufacturing process. Many process patents also include examples based on computer simulations, in which case the data should be viewed with suspicion. When using data from patents, it is important to carefully read the section that describes the experimental procedure to be sure that the experiments were run under appropriate conditions.

A patent gives its owner the right to sue anyone who practices the technology described in the patent claims without a license from the patent owner. Patent attorneys generally try to write patents to claim broad ranges of process conditions so as to maximize the range of validity and make it hard for competitors to avoid the patent by making a slight change in temperature, pressure, or other process parameters. Very often, a patent will say something along the lines of “the reaction is carried out at a temperature in the range 50° to 500 °C, more preferably in the range 100° to 300 °C and most preferably in the range 200° to 250 °C.” It is usually possible to use engineering judgment to determine the optimal conditions from such ranges. The best conditions will usually be at or near the upper or lower end of the narrowest defined range. The examples in the patent will often indicate the best operating point.

Patents can be downloaded for free from the website of the U.S. Patent Office, www.uspto.gov. The U.S. PTO website also has limited search capability. The entire U.S. PTO collection is also available at www.google.com/patents. Most large companies subscribe to more sophisticated patent search services such as Derwent Innovation (www.derwentinnovation.com), PatBase (www.patbase.com), or GetthePatent (www.getthepatent.com).

Several guides have been written to help engineers understand the use of patents for the protection of inventions and as sources of information; such as those by Auger (1992) and Gordon et al. (2017).

Consultants

Engineers in industry often hire specialist consulting firms to prepare analyses of commercial technology. Consultants can be used to provide an impartial assessment of a competitor's or vendor's process. Some consulting firms such as SRI and Nexant regularly publish assessments of the technology available for making different chemicals. These assessments are based on flowsheets and design models that the consultants have developed from information that they gathered from the literature and from direct contact with the technology suppliers.

Some caution is needed when working with consultants. The client must carry out due diligence to ensure that the consultant is truly impartial and does not bias their analysis. The client should also cross-check the information provided by the consultant against recent patents and publications to ensure that the consultant is working from the latest information.

Vendors

Technology suppliers and contractors will sometimes make design information available to a client in the hope of securing a sale. If a project team requires information to make a technology selection, the technology vendors may be willing to supply edited PFDs (for example, with stream flows blanked out or with some information missing), reactor yields, or even designs from a similar plant at a smaller production scale. More detailed information is usually provided by vendors when bidding on a contract for a project that has a high likelihood of going forward.

2.4.2 Factors considered in process selection

Once the design team has assembled information on the alternative commercial processes, they will usually need to carry out substantial customization and optimization of the designs before a selection can be made.

The information given in the open literature is usually restricted to block flow diagrams and (occasionally) reactor yields. The first step is usually to complete a full PFD and mass and energy balance of the process. These can be used for preliminary sizing and costing of the main process equipment to obtain an estimate of the required capital investment, as described in Chapter 7. The feed and product flow rates and energy consumption can be used to estimate the costs of production, as described in Chapter 8. The economic analysis methods introduced in Chapter 9

can then be applied to determine the overall project economics and choose which design gives the best overall economic performance according to the criteria established by the company.

If one process flowsheet has a particular cost advantage, this will usually become clear in the economic analysis. Factors such as feedstock or fixed-cost advantages that can be very important in selecting *between projects* are usually less important when selecting *between flowsheets* within a given project. The selection between flowsheets is usually influenced more by process yields, energy consumption, and capital requirements, and hence is sensitive to catalyst, organism or enzyme performance, and process design and optimization.

In an industrial context, technology vendors or engineering, procurement, and construction (EPC) contractors will often supply detailed PFDs and material and energy balances to a client when invited to bid on a project. Some diligence is always needed in checking the information in proposals and verifying performance claims against the actual performance of recent plants built by the vendor.

Although an economic analysis is always carried out, it is usually not the sole criterion for technology selection. Some other important factors are described next.

Freedom to practice

Freedom to practice is a legal concept that arises from patent law. If a process, catalyst, enzyme, genetically modified organism, or chemical route is patented, it can only legally be used under license from the patent holder. If another company were to use the technology without a license, they would be infringing on the patent, and the patent holder could sue to stop the use and demand damages.

Determination of freedom to practice usually requires the expert advice of patent attorneys. In rapidly evolving new fields it can be difficult to assess, as patent applications typically are not published until 1 to 2 years after they are filed, so a decision may be made to proceed with building a plant before the intellectual property space can be completely mapped. Another complication is that competing technology vendors may have overlapping patents or patents that appear to block features of each other's designs.

All patents are only valid for a fixed term; in the United States at the time of writing this is 20 years from the date the patent was filed. When a patent has expired, anyone is free to practice the technology. Care must still be taken to check that the original technology developer has not made more recent improvements that are still under patent protection. Many conventional processes are no longer protected by patents and can be bought from EPC companies without paying a royalty or license fee.

When a customer licenses technology from a vendor, the vendor will usually indemnify the customer against patent infringement. This means the technology vendor asserts that they have ownership of the technology and freedom to practice and that they will help the customer fight any patent infringement suits brought by their competitors. Technology vendors sometimes minimize the potential for such lawsuits by forming cross-licensing agreements.

Safety and environmental performance

All commercially practiced technologies should meet or exceed the minimum legally acceptable safety standards, but some older processes may no longer have acceptable environmental performance.

An economic analysis will usually not distinguish whether one process is safer or more environmentally acceptable than another. The methods described in [Chapters 10 and 11](#) can be used to make an assessment of process safety and environmental impact.

When assessing commercial technology, visits to existing sites and reviews of their safety and operational performance can also be helpful.

Government and international restrictions

Governments sometimes place restrictions on companies that can influence technology selection. It is fairly common for nationally owned companies in developing countries to be required to maximize use of indigenous technology, equipment, and parts so as to stimulate the development of local engineering industries and reduce hard currency outflows. This may lead a company to develop its own version of an older technology rather than working with a technology vendor or major international company that can supply the latest technology.

International sanctions can also play an important role in process flowsheet selection. Sanctions can disqualify some companies from offering to supply technology and reduce the set of options available. Sanctions can also restrict the availability of feedstocks. During the 1970s and 1980s, South African companies developed many processes for making chemicals from coal in response to the international sanctions aimed at ending apartheid that restricted their ability to purchase crude oil.

Experience and reliability

One of the critical factors in selecting a commercially proven technology is the extent and diversity of operating experience that has been established. If a process has been widely adopted and proven in many locations by different operating companies, then it is likely to be easy to apply in a new plant. A technology that has only been built once or twice may still experience “teething troubles” and be more difficult to implement.

As more operating experience is gained, the company also gains a better understanding of the reliability of the process. If a particular section of the flowsheet or piece of equipment is known to cause reliability problems, this may create a need to modify the equipment design or even make changes to the flowsheet.

2.4.3 Modification and improvement of established processes

All designs evolve over time. Engineers make modifications to improve process economics, safety, reliability, and environmental impact. Most changes will be minor, such as addition of instrumentation or substitution of equipment; however, significant changes in the flowsheet are sometimes needed.

Modifications that are made to an existing commercial plant are known as *revamp designs* and are addressed in Section 2.5. This section describes techniques for modifying an established process for use in a new plant.

Modifications to improve process economics

Improvements in process economics usually come from reduced capital investment or improved cost of production. Designers seeking improvements in process economics usually start by completing a PFD of the existing design and determining the current estimated capital investment and cost of production (see [Chapters 7, 8, and 9](#)). The following tactics can then be applied:

- **Improve reactor selectivity and process yield.** Feedstock costs are usually more than 80% of the cost of production, so improving yields gives the biggest impact on process economics. Improved yields usually require the development of more selective catalysts, enzymes, or organisms, or a more effective reactor design, but sometimes a more efficient separation scheme or better purification of a feed or recycle will also improve yield.
- **Improve process energy efficiency.** Energy costs are usually next largest after feedstock costs for chemicals produced on a large scale. Energy costs can be reduced by improving process energy efficiency. Several different approaches to improving process energy use are described in [Chapter 3](#).
- **Improve process fixed costs.** Fixed costs are usually second to feedstock costs in small-scale processes used for fine chemicals and pharmaceuticals manufacture. Fixed costs are described in Section 8.5. Fixed costs can be reduced by making the process more continuous and less labor-intensive and by increasing the plant attainment of batch processes.
- **Reduce capital investment.** Design engineers look for pieces of equipment that can be combined or eliminated to reduce capital cost. In batch plants, this is often done by carrying out several steps in the same piece of equipment. For example, the feed can be charged to a reactor, heated in the reactor, reacted, cooled down, and the product crystallized before pumping out the product as a slurry and repeating the process.
- **Reduce working capital.** Working capital is described in Section 9.2.3. Working capital can be reduced by decreasing inventories of raw materials, work in progress, and consumables. Making a process more continuous and using fewer different solvents in a process both lead to a reduction in working capital.

It can be seen that some of these suggestions contradict each other; for example, “make batch plants more continuous” but “carry out more operations in the same piece of equipment.” Lists of design guidelines (known as *heuristic rules*) often contain apparent contradictions. The designer must either choose which rule is most appropriate to the case under consideration using experience and judgment or else carry out a full design and costing of both alternatives. Heuristic rules are discussed further in the context of process synthesis in Section 2.6.4.

Modifications to improve plant safety

Plants can be made more inherently safe by reducing inventories of hazardous materials by making vessels and other plant equipment smaller; substituting less hazardous materials for feeds, solvents, and intermediates; eliminating explosive mixtures and exothermic reactions; eliminating use of operations that are open to the atmosphere; minimizing worker exposure to chemicals; and other methods discussed in [Chapter 10](#).

Improvements in the safety of a design can be quantified using the methods for risk assessment described in Section 10.8.

Modifications to improve plant reliability

When a plant has been in operation for a few years the operators will have a good idea of which plant sections or pieces of equipment lead to the most operational problems, require the most maintenance, and cause the most unplanned shutdowns.

Reliability problems are usually caused by equipment failures. The most common problems are usually experienced with solids-handling equipment, rotating equipment such as pumps and compressors, heat exchangers that are prone to fouling, and instruments and valves. Sometimes, specification of a more reliable piece of equipment can solve a process reliability problem. More often, a flowsheet change is needed, such as designing with two or more pieces of equipment in parallel so that the plant can continue to operate while one is taken offline for repair or cleaning. This approach is commonly applied for pumps, which are relatively cheap and prone to stalling in operation.

Corrosion, erosion, and plugging caused by corrosion products can be major contributors to poor reliability. Methods to address corrosion in design are described in Chapter 6.

Modifications to improve environmental impact

Many conventional processes were first designed over 40 years ago, when different environmental laws and standards applied. Existing plants may have been modified by addition of end-of-pipe systems for reducing environmental impact; however, changes in the process flowsheet can sometimes achieve the same or better environmental performance at lower cost.

Modifications that are typically used to improve environmental impact include:

- Use of new catalysts, enzymes, or organisms that have better selectivity for the desired product and consequently lead to less waste formation.
- Optimization of reactor design to give better mixing or heat transfer and hence improve reactor selectivity and reduce by-product formation.
- Elimination of solvents or other consumables that become degraded to waste products by the process.
- Elimination of materials that have high environmental impact, such as halogenated solvents, mercury, endocrine disruptors, and compounds that persist in the environment.
- Adoption of closed-loop recirculating gas systems instead of once-through gas flow, thus reducing volatile organic compound (VOC) emissions. For example, Fig. 2.13(a) shows a once-through dryer in which the drying gas is vented or sent to a flare, potentially leading to VOC emissions. Fig. 2.13(b) shows an alternative design in which a blower is used to circulate the gas. The hot gas leaving the dryer is cooled to allow solvent to be condensed and recovered. There is much less potential for VOC emissions in the closed-loop design, and the consumption of solvent is also reduced.
- Substitution of chemicals with materials that have reduced environmental impact. For example, the cheapest way to neutralize waste sulfuric acid is to react it with lime (CaO) to form gypsum (CaSO_4), which is inert and can be sent to a landfill. Instead, if ammonia is used to neutralize the acid, the product will be ammonium sulfate, which can be used as a fertilizer.

Methods for analyzing and reducing the environmental impact of a process are described in Chapter 11.

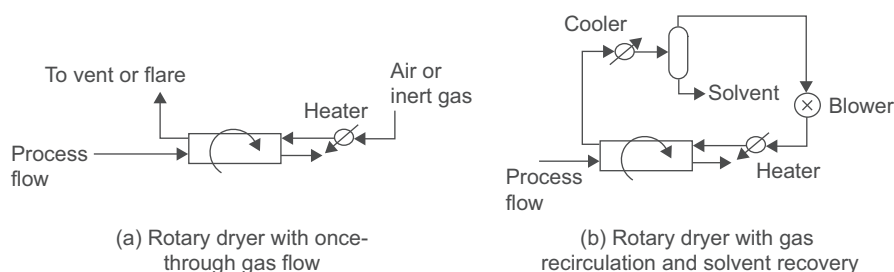


FIG. 2.13 Dryer gas circulation designs.

2.5 Revamps of existing plants

Flowsheet development for plant revamps is a specialized subject in its own right. Revamp design is rarely taught in universities, as revamp studies require access to an operating plant and the data it produces.

Revamps generally fall into two categories. *Debottlenecking* projects are carried out to increase the production rate of a plant while making the same product. *Retrofit* projects are carried out to change the design of a plant to handle different feeds; make different products; exploit better reactor, catalyst, or separation technology; or improve plant safety or environmental impact in response to new regulatory requirements.

2.5.1 Flowsheet development in revamp projects

Fig. 2.14 gives an overall work process for developing a revamp design flowsheet. One of the critical requirements of a revamp project is always to minimize project cost by maximizing reuse of existing equipment. The revamped flowsheet therefore always requires compromises between desired objectives and what can be obtained with the equipment available.

Many features of a revamped flowsheet will be different from the flowsheet of a corresponding new plant. For example, in a revamp it may make sense to add a second distillation column in parallel to an existing column rather than tearing down the existing column and building a new larger one. The use of two small columns in parallel would not be contemplated in a new design. The development of a revamped flowsheet thus requires a lot of information on the performance of existing equipment so that the equipment can be rerated or modified for a role in the new flowsheet. When the existing equipment cannot be upgraded, the designer must find the cheapest method to add new capacity or augment the existing capacity.

Once the revamped flowsheet has been completed, the designers can assess the costs of the new components that must be added. The cost of revamping a plant should always be compared with the cost of building a new plant from scratch. The revamp will usually be a cheaper method for adding small increments of capacity, but for larger capacity increases, a new unit will become more attractive.

The steps in developing a revamped process flowsheet are described in the following sections. Although there is a great deal of retrofit and revamp activity in the chemical industry, particularly in regions such as the United States

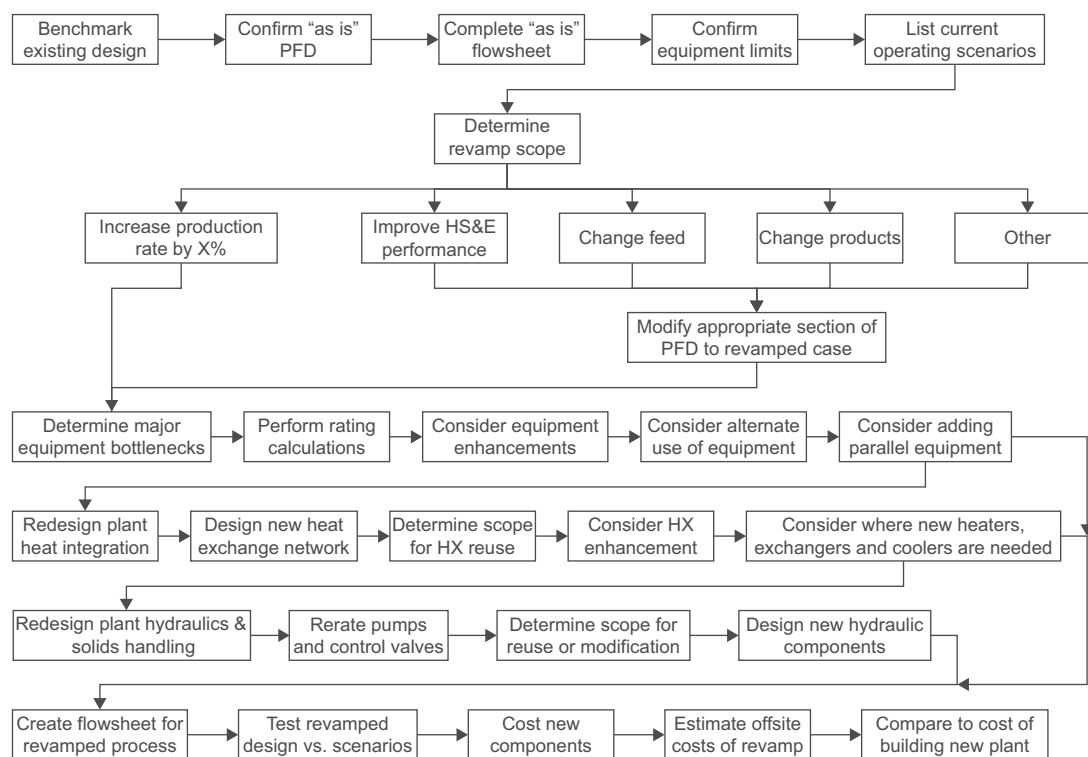


FIG. 2.14 Steps in revamp design.

and European Union, where the industry has been long-established, the authors are not aware of any comprehensive reference works on the subject. The books by Briggs et al. (1997) and Douglas (1988) contain short sections on revamp design. Process revamps for energy savings are discussed in detail by Zhu (2014).

2.5.2 Major equipment debottlenecking

In a revamp design, the capacity of existing equipment determines whether additional equipment must be added in series or parallel, and hence plays a major role in determining the revamped design flowsheet.

Most major equipment is initially specified with a design factor or margin of 10% to 20%; see Section 1.6. This overdesign allows for errors in the design data and methods, but also creates some room for potential expansion of capacity. When a plant is considered for revamp, some of the equipment may still be operating below its full capacity.

The general procedure for equipment debottlenecking follows the steps shown in Fig. 2.14. Once a mass and energy balance has been established for the existing plant, a simulation model of the equipment can be built. The model can then be tested under the proposed new process conditions to determine if the equipment is fit for the new service. For equipment that is difficult to model (for example, centrifuges, fired heaters, and dryers) a specialist or the original equipment vendor may need to be consulted. After establishing the maximum capacity that the equipment can attain while maintaining specifications, modifications to improve capacity can be considered. If it is not possible to satisfy the desired process duty with modifications to the existing equipment, then the cheapest means of adding capacity must be established. This may include complete replacement of the original equipment, with reuse of the original equipment elsewhere in the process.

Some specific examples of techniques for equipment debottlenecking are given next. Revamp of heat transfer equipment is discussed in Section 2.5.3, and revamp of hydraulic and solids handling equipment is described in Section 2.5.4.

Reactor debottlenecking

Reactors are designed with a specified residence time that has been determined to give a desired conversion. For reactors that use a fixed bed of catalyst, this is usually expressed instead as a space velocity:

$$\tau = \frac{V}{v} \quad (2.8)$$

$$SV = \frac{v}{V_{cat}} \quad (2.9)$$

where: τ = residence time

SV = space velocity

V = reactor volume

V_{cat} = fixed bed catalyst volume

v = volumetric flow rate

Space velocity is usually given on an hourly basis and defined on the basis of gas phase flow (GHSV = gas hourly space velocity), liquid phase flow (LHSV = liquid hourly space velocity), or total mass flow (WHSV = weight hourly space velocity = kg/h feed per kg of catalyst inventory). Any consistent set of units can be used for residence time and space velocity.

Equations 2.8 and 2.9 clearly show that an increase in flow rate must lead to a proportional change in either volume, residence time, or space velocity. Adding volume usually requires building additional reactors, so unless the reactors are very inexpensive, the revamp design will focus on ways to reduce residence time or increase space velocity while trying to maintain the same conversion, if possible, so as to minimize the impact on the separation and recycle sections of the plant.

An increase in space velocity or decrease in residence time can be obtained by increasing temperature; backing off on conversion; reducing the concentration of diluents; or using a more active catalyst, enzyme, or organism. Increasing temperature and reducing diluents or solvents (if any are present) will generally lead to worse selectivity and increased cost elsewhere in the process. Many fixed-bed catalytic processes are operated on a temperature cycle, where the reactor temperature is slowly increased over a 1- to 10-year period to compensate for catalyst deactivation and the catalyst is then replaced at the end of the cycle. In such cases, raising the temperature shortens the catalyst run length and requires more frequent plant shutdown. Reducing the reactor conversion also creates additional cost

elsewhere in the process as the amount of unreacted feed recycle is increased. Improving the catalyst performance is often the least expensive way to boost capacity, and the availability of new catalysts often sets the scope for revamp projects.

An additional problem with revamping fixed-bed catalytic reactors is the effect of reactor pressure drop. The pressure drop across a packed bed is proportional to the flow rate squared, so pressure drop increases rapidly as flow rate is increased. Approaches that can be taken to reduce reactor pressure drop include rearranging series reactors into parallel flow (Fig. 2.15) and converting down-flow reactors to radial flow (Fig. 2.16). (A more detailed drawing of a radial flow reactor is given in Chapter 15; see Fig. 15.29). With packed beds that are in up-flow, care must be taken to avoid fluidizing the catalyst in the revamped design. If the up-flow velocity is close to the minimum fluidization velocity, then the reactor should be converted to down-flow or replaced with a larger reactor. Sizing of packed bed reactors is discussed in more detail in Section 15.7.3.

When it is necessary to add reactor capacity, a technique that is widely used is to add a pre-reactor to the existing reactor sequence, as in Fig. 2.17. Because a pre-reactor typically runs at low conversion, it can be operated under conditions that would not normally be good for selectivity; for example, at higher temperature or with less solvent or diluent. This makes the pre-reactor more volume-efficient than the existing reactor sequence, without compromising on overall selectivity.

When a detailed model of the reaction kinetics, including side reactions, is available, more complex reactor networks can be designed that give improved selectivity and yield of desired products. A revamp project can then add reactor capacity to bring the reactor section performance closer to that of the ideal reactor network. The design of reactors and reactor networks is discussed in more detail in Chapter 15.

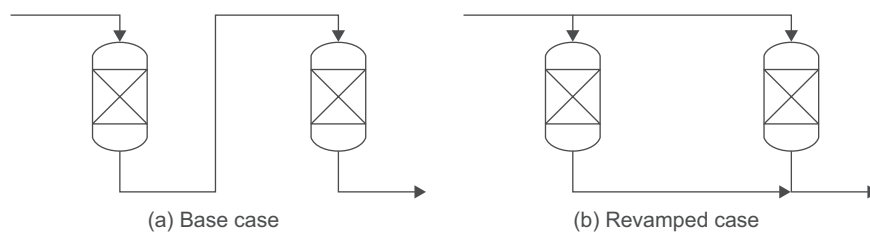


FIG. 2.15 Series to parallel reactor revamp.

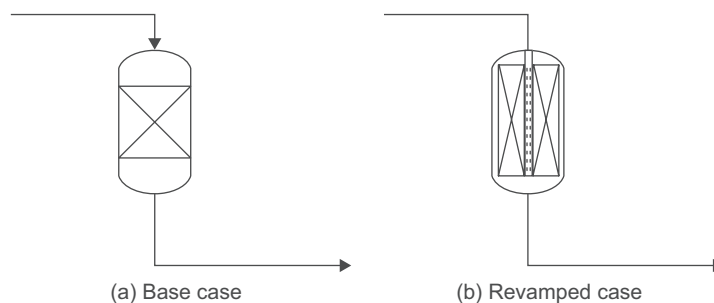


FIG. 2.16 Down-flow to radial-flow reactor revamp.

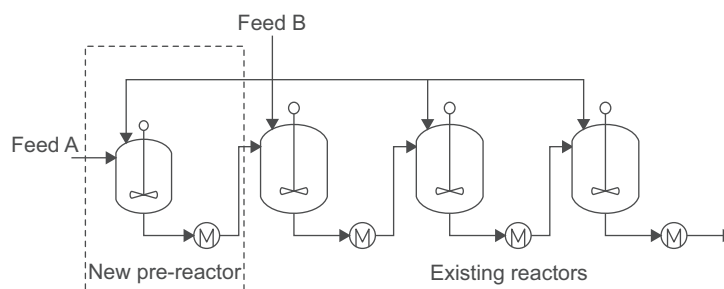


FIG. 2.17 Reactor revamp using a pre-reactor.

Separation column debottlenecking

The capacity of separation columns is usually limited by column hydraulics; see Section 17.13. If the feed flow rate increases, the vapor rate in the column increases proportionately, and at some point the column will flood and become inoperable. Two approaches can be taken to obtain more capacity:

1. Increase the open area for vapor flow to delay the onset of flooding.
2. Increase the number of stages in the column by using high-efficiency trays or trays that allow a closer tray spacing, so that the reflux ratio, and hence vapor rate, can be reduced.

Both of these methods are used by separation tray and packing vendors, and there are many proprietary designs of high-efficiency, high-capacity trays and packing on the market. When revamping a column, the common practice is to contact the tray and packing vendors, who will then supply an estimate of how many trays must be replaced to achieve a desired capacity. It is often not necessary to re-tray the entire column. The detailed design of distillation columns is discussed in [Chapter 17](#).

When it is necessary to add capacity to a distillation column and the use of high-capacity trays is not sufficient, a prefractionator scheme is sometimes used, as illustrated in [Fig. 2.18](#). The prefractionator makes a preliminary separation of the feed that reduces the reflux requirements of the main column. A revamped prefractionator is usually provided with its own reboiler and condenser to avoid increasing the load on the main column reboiler and condenser.

Another common tactic in debottlenecking separation sections is to reuse the existing distillation columns in a different application. If a plant has three or more columns, then a revamp can be carried out by building one new column to replace the largest of the existing columns, revamping the largest old column to replace the second largest, etc. In this context, largest refers to the largest diameter, which governs vapor rate and capacity. If the columns do not have sufficient height for the new application, they can be re-trayed with high-efficiency internals or combined in series. In some cases, particularly for relatively short low-pressure columns, it may even make sense to add height to a column by welding on a new top section.

2.5.3 Revamp of heat exchange networks

The heat exchangers, heaters, and coolers of a plant will cause many of the bottlenecks to plant expansion. When a plant is revamped to a new purpose, such as changed feed or products, the existing heat recovery system will no longer be optimal and may no longer make good sense. The common practice in major revamps is to complete the revamp design of the other major equipment first, then address the heaters, coolers, and exchangers subsequently, as shown in [Fig. 2.14](#).

The design of heat recovery systems and heat-exchanger networks is covered in [Chapter 3](#). [Chapter 19](#) addresses the detailed design of heat exchangers, heaters, and coolers. Revamp of a complex heat-exchange system should

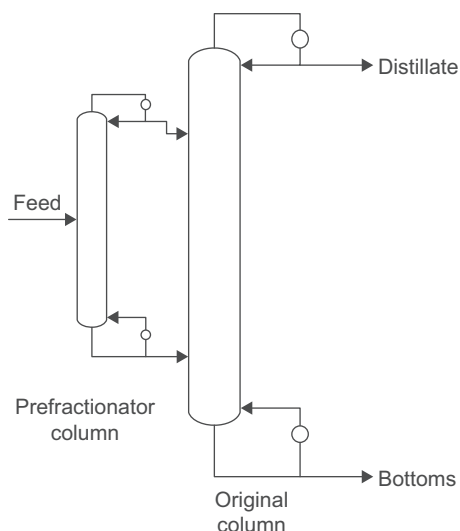


FIG. 2.18 Distillation column revamp using a prefractionator.

always be treated as a network problem rather than by revamping each item individually, as the lowest-cost solution will always be that obtained by optimizing the system as a whole.

The revamp of heat-exchanger networks has been the subject of much research, and very effective techniques and software for heat-exchange network revamp have been developed. The *network pinch* method developed by [Asante and Zhu \(1997\)](#) is now the most widely used method in the industry. This method has been automated ([Zhu and Asante, 1999](#)) and is used by most of the companies that offer heat integration consulting services. [Smith \(2016\)](#) gives a concise overview of the network pinch approach; see also [Zhu \(2014\)](#).

For simple processes with only a few heaters and coolers, the tactics described in the following sections can be used.

Heat exchangers

For a heat exchanger:

$$Q = UA\Delta T_m \quad (2.10)$$

and

$$Q = m_i C_{p,i} \Delta T_i \quad (2.11)$$

where : Q = heat transferred per unit time, W

U = the overall heat-transfer coefficient, W/m^2K

A = heat-transfer area, m^2

ΔT_m = the mean temperature difference, the temperature driving force, $^{\circ}C$

m_i = the mass flow rate of stream i , kg/s

$C_{p,i}$ = the specific heat capacity of stream i , J/kgK

ΔT_i = the change in temperature of stream i for a stream that undergoes only sensible heat changes, $^{\circ}C$

Increasing the flow rate increases the required duty and therefore requires an increase in heat transfer coefficient, area, or effective temperature difference.

The correlations that are used to predict heat transfer coefficients for sensible heat transfer are usually proportional to $Re^{0.8}$, where Re is the Reynolds number, which is proportional to flow rate. Hence, the process-side transfer coefficient increases almost in ratio to the increase in flow unless the exchanger is boiling or condensing the process stream. For heaters and coolers, it may therefore be possible to reuse the exchanger in the same service if an increase in utility-side heat transfer coefficient or a change in utility temperature can make up the rest of the required duty. This is topic explored in Example 2.7.

One of the first and most important steps in revamping a heat exchange system is to benchmark the current system and estimate the heat transfer coefficients that are currently being obtained. If these are substantially lower than expected from the original design of the process, this may indicate fouling, plugging, or other problems that should be addressed during the revamp.

There are several proprietary methods for enhancing the performance of tubular exchangers. Tube inserts such as hiTRAN, TURBOTAL, and Spirelf can be used to increase turbulence and tube-side heat transfer coefficient. Low-fin tubes can be used to increase shell-side effective area ([Wolverine, 1984](#); see also Section 19.14). Reboiling and condensing coefficients can be increased by use of UOP High Flux or High Cond tubing.

Plate exchangers usually do not require enhancement methods. Gasketed plate exchangers are very easy to revamp, as more plates can simply be added to the exchanger; see Section 19.12. Welded plate exchangers are not amenable to the use of inserts or to expansion by adding plates.

Heaters and coolers

The techniques described for heat exchangers apply equally well to steam or oil heaters and water coolers. Decreasing cooling water return temperature (by increasing cooling water flow rate) and raising hot oil temperature are widely used methods in plant revamps.

Fired heaters are usually difficult to revamp and require the involvement of heater and burner specialists. If there is space in the heater, additional tubes can be added. Similarly, if there is space in the convective section, it can be used for pre-heat to off-load some of the furnace duty. In some cases, the addition of improved burners will allow more uniform heating and higher average tube-wall heat flux. Fired heater design is discussed in more detail in Section 19.17.

Air coolers (Section 19.16) are also difficult to revamp. The designer does not have the ability to specify a lower ambient temperature! Common air cooler revamps include:

- Adding more banks of tubes and installing more powerful fans.
- Adding water-spray systems to increase cooling capacity on the hottest days.
- Water-spray systems are effective, but can increase air-side fouling over time.

Example 2.7

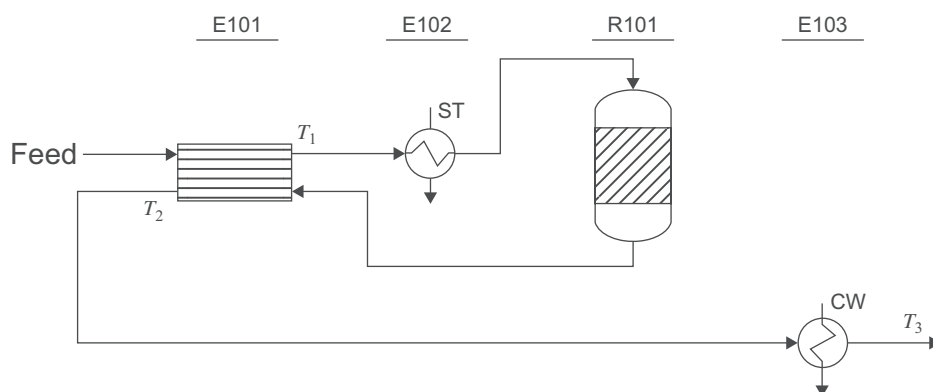
Fig. 2.19 shows a simple heat-exchange system. A feed stream is heated by heat exchange in a plate exchanger and then further heated in a steam heater before entering a fixed-bed reactor. The product from the reactor is cooled in the plate exchanger and then further cooled using cooling water. Exchanger specifications and current performance are given in the figure. Propose modifications to the system to allow a revamp to 50% higher capacity.

Solution

Start with the plate heat exchanger, E101. A 50% increase in flow through the exchanger would give a $(1.5)^2 = 2.25$ factor increase in exchanger pressure drop. If this was acceptable, then assuming that the heat transfer coefficient is proportional to $Re^{0.8}$:

$$\text{New heat transfer coefficient} = 350 \times (1.5)^{0.8} = 484 \text{ W/m}^2\text{K}$$

Assuming the reactor operates at the same outlet temperature, we can make heat balances for the base case and revamp case.



Exchanger	E101	E102	E103
Type	Plate	S&T	S&T
Duty (kW)	800	400	200
Hot side T in ($^{\circ}\text{C}$)	140	180	60
Hot side T out ($^{\circ}\text{C}$)	60	180	40
Cold side T in ($^{\circ}\text{C}$)	40	120	25
Cold side T out ($^{\circ}\text{C}$)	120	160	35
F_r factor	1	1	0.92
Hot side heat transfer coefficient ($\text{W/m}^2\text{K}$)	700	2000	700
Cold side heat transfer coefficient ($\text{W/m}^2\text{K}$)	700	500	700
Overall heat transfer coefficient ($\text{W/m}^2\text{K}$)	350	400	350
Area (m^2)	114	27.5	31.7
ΔT_{lm}	20	36.4	19.6

FIG. 2.19 Heat-exchange system for Example 2.7.

Base case:

$$Q_{101} = 800 \times 10^3 = m_f C_{p,f} (120 - 40) = m_p C_{p,p} (140 - 60)$$

$$m_f C_{p,f} = m_p C_{p,p} = 10^4$$

Where subscript f denotes feed and p denotes product.

Revamp case:

$$Q_{101} = 1.5 m_f C_{p,f} (T_1 - 40) = 1.5 m_p C_{p,p} (140 - T_2)$$

$$T_1 = 180 - T_2$$

and

$$Q_{101} = UA \Delta T_m = 484 \times 114 \times (140 - T_1)$$

So

$$1.5 \times 10^4 (T_1 - 40) = 484 \times 114 \times (140 - T_1)$$

hence

$$T_1 = 118.6^\circ \text{C}$$

$$T_2 = 61.4^\circ \text{C}$$

So we only lose 1.4 degrees of heat exchange as long as the increase in pressure drop that results from the higher flow rate is acceptable.

Now look at the steam heater, E102. In the revamp case:

$$Q_{102} = 1.5 \times 10^4 (160 - 118.6) = 621 \text{ kW}$$

The heat transfer coefficient is only increased on the process (cold) side:

$$\text{New cold - side coefficient} = 500 \times (1.5)^{0.8} = 691.6 \text{ W/m}^2\text{K}$$

$$\text{New overall heat transfer coefficient} = \left((2000)^{-1} + (691.6)^{-1} \right)^{-1} = 513 \text{ W/m}^2\text{K}$$

(Overall heat transfer coefficient calculated using Equation 19.2)

So for the exchanger to be feasible, we would need to raise the steam temperature to T_s , where:

$$Q_{102} = UA \Delta T_m$$

$$621 \times 10^3 = 513 \times 27.5 \times \left(\frac{(T_s - 160) - (T_s - 118.6)}{\ln \left(\frac{T_s - 160}{T_s - 118.6} \right)} \right)$$

$$T_s = 186.5^\circ \text{C}$$

So the heater remains viable if the steam temperature can be increased by 6.5°C . This might be accomplished by raising the local pressure of the medium pressure steam system, for example, by making adjustments in the steam pressure regulator valve set points. If that was not possible, and if the heater was rated for a sufficiently high pressure, then high-pressure steam could be considered instead of medium-pressure steam.

Turning to the cooler, E103. In the revamp case:

$$Q_{103} = 1.5 \times 10^4 (61.4 - 40) = 321 \text{ kW}$$

To meet this extra duty we need to increase the cooling water flow rate by a factor F_{cw} , which changes the cooling water outlet temperature to T_w , where:

$$Q_{103} = 321 \times 10^3 = F_{cw} \times 20 \times 10^3 (T_w - 25)$$

and

$$321 \times 10^3 = UAF \Delta T_{lm}$$

For the revamp case, the logarithmic mean temperature difference ΔT_{lm} depends on T_w and the overall heat transfer coefficient will be:

$$\frac{1}{U} = \frac{1}{700 \times (1.5)^{0.8}} + \frac{1}{700 \times (F_{cw})^{0.8}}$$

These equations must be solved iteratively. This is easily done using a spreadsheet, giving:

$$\begin{aligned} F_{cw} &= 1.89 \\ T_w &= 33.5^\circ \text{C} \end{aligned}$$

The exchanger would be feasible if we could accept this increase in cooling water flow rate; however, such a large increase would give a very large increase in pressure drop. The pressure drop is proportional to flow rate squared and so would increase by a factor $(1.89)^2 = 3.57$, which would probably not be acceptable.

An alternative approach for E103 is to see what the outlet temperature would be if cooling water flow were limited. Suppose the cooling water flow cannot be increased more than 20%, giving a 44% increase in pressure drop. If we allow the cold outlet temperature and hot outlet temperature to vary, the same spreadsheet model can be solved to give:

$$\begin{aligned} T_w &= 36.4^\circ \text{C} \\ T_3 &= 43.1^\circ \text{C} \end{aligned}$$

So in this case the designer would have to consider whether the product stream could be made 3.1°C hotter, which depends on the downstream processing.

If this option was also unacceptable, the designer would have to consider adding another cooler. A simple approach would be to split the hot stream, leaving E101 into two streams in a ratio of 2:1. The larger stream could still be sent to E103, and the smaller one could be sent to a parallel cooler half the size of E103. This option would not push the operation of E103 much beyond the current operating mode.

In summary, one option that would allow a 50% increase in throughput would be:

- **E101:** No change, as long as the factor 2.25 increase in pressure drop is acceptable on hydraulic review.
- **E102:** Increase steam temperature to 186.5°C ; no capital modification is needed.
- **E103:** Consider accepting a 3.1°C warmer product. If this is unacceptable, add a new exchanger E104, half the size of E103, in parallel to E103 and split the hot stream, leaving E101 in a 2:1 ratio between E103 and E104.

Alternatively, if E101 is a gasketed plate heat exchanger (see Section 19.12.1), the designer could also consider adding more plates to increase the area available in E101. We can increase the plates in E101 until we achieve the same channel velocity and pressure drop as the base case then solve the remaining problem for E102 and E103, or, alternatively, increase the plates in E101 until the existing E102 and E103 are able to satisfy all of the remaining heating and cooling load without modifications. This alternative solution is explored in Problem 2.11. The addition of plates to E101 would be more expensive than just altering a few temperatures, but would most likely be less costly than adding E104 and could even be cheaper than making the hydraulic modifications necessary to cope with a large increase in pressure drop.

2.5.4 Revamp of plant hydraulics

Any revamp project that leads to increases in plant throughput will have a significant effect on the plant hydraulic equipment. Because pressure drop is proportional to velocity (and hence flow rate) squared, a 40% increase in flow is sufficient to double the pressure drop. The introduction of parallel equipment in the flowsheet may also create a need for additional control valves to regulate flow in the desired ratios, which adds more pressure drop to the plant. Adding equipment to modify the process also has an impact on the hydraulic design.

Much of the design effort in revamp projects goes into evaluating and redesigning the plant hydraulics. All of the pump-and-line calculations and control valve sizing calculations must be repeated for the revamped design case. It is usually not cost-effective to replace the piping with new pipe of more optimal diameter, so usually the designers will accept a higher pressure drop in the pipes and process equipment and then redesign the pumps and control valves accordingly.

The design of pumps, compressors, piping systems, and control valves is discussed in more detail in [Chapter 20](#). The following sections provide some specific guidelines relevant to revamp flowsheet development.

Compressors

Compressors are the largest and most expensive items in the plant hydraulic equipment. Compressor design is covered in more detail in Section 20.6. Because compressors are expensive to replace, experienced designers usually try to reuse the existing compressors in the new flowsheet.

The relationship between flow rate and pressure delivered depends on the type of compressor (Fig. 2.20), but the pressure delivered will usually decrease if the compressor operates at higher flow. The only exception to this rule is large reciprocating compressors, which are usually designed with a recycle from the product to the feed, known as a *spill-back*. If the spill-back flow is large enough, some increase in flow can be obtained by reducing the spill-back, with no loss in delivered pressure.

When two compressors are used in parallel, the total flow increases, but the delivered pressure cannot be greater than the lower pressure delivered by either compressor; see Fig. 2.21. When two compressors are used in series, the pressure delivered increases, but the flow rate does not; see Fig. 2.22.

In a revamp flowsheet design, the designer seeks to increase the pressure delivered to overcome the increased plant pressure drop, as well as an increase in flow rate. From Figs. 2.21 and 2.22 it can be seen that the only way to increase both the pressure delivered *and* the flow rate while adding only one compressor is somewhat paradoxically to *reduce* the flow rate in the existing compressor so that it delivers a higher pressure and then add a second machine in parallel, as shown in Fig. 2.23. The extent to which the flow rate can be reduced depends on the type of compressor and current operating conditions. In this scenario, the revamped flowsheet would need to show two

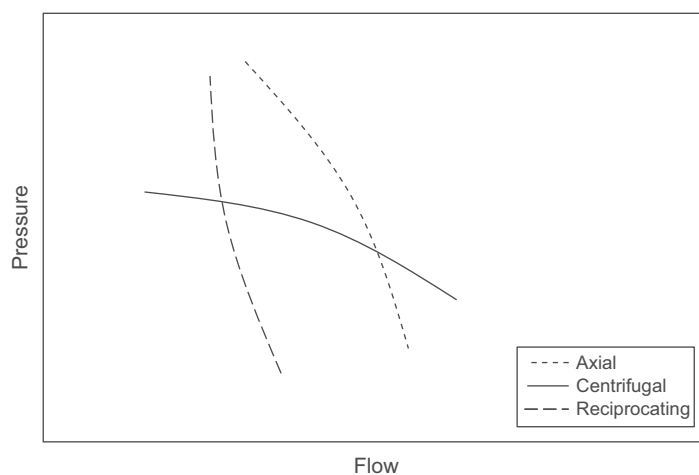


FIG. 2.20 Pressure-flow rate curves for different compressor types.

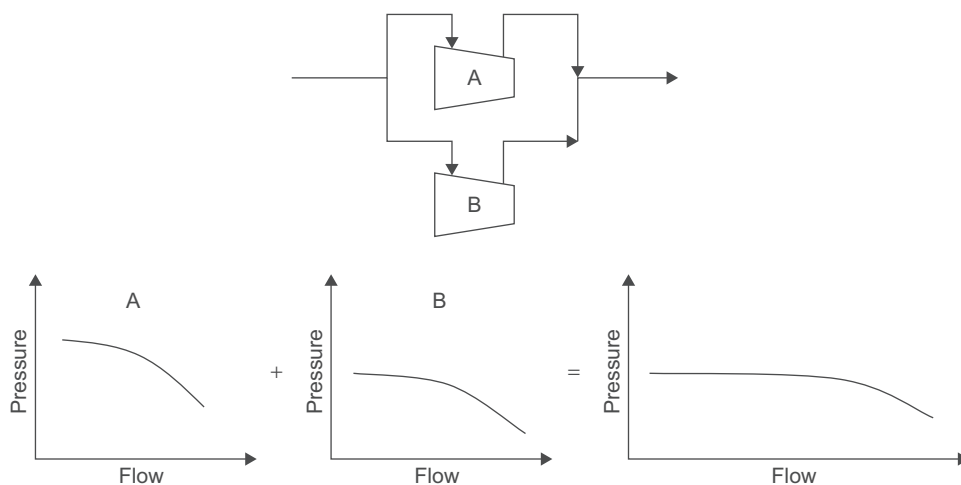


FIG. 2.21 Compressors in parallel.

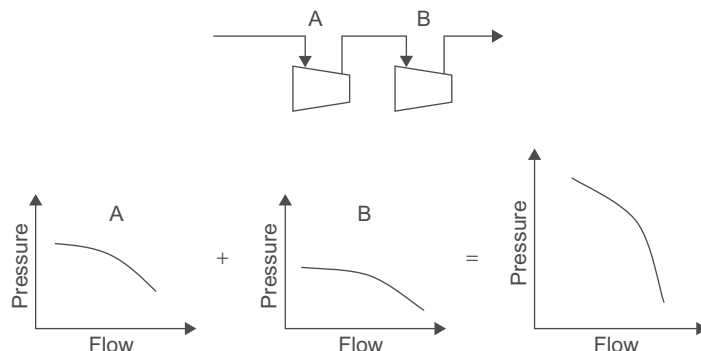


FIG. 2.22 Compressors in series.

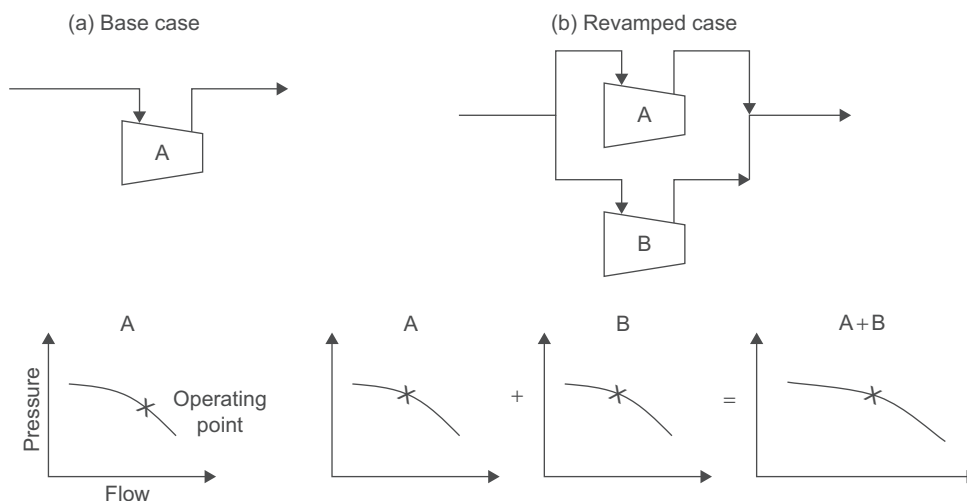


FIG. 2.23 Compressor revamp.

compressors in parallel. A revamp from one machine operating at 100% flow rate to two similar machines in parallel, each at 70% flow rate, would give an overall 40% increase in flow.

If it is not possible to meet the desired flow and pressure with the addition of only one new compressor, it will usually be preferable to replace the existing compressor rather than adding new compressors in series and parallel.

If a compressor cannot be reused in its current location in the revamped flowsheet, it should be evaluated for other process uses. Air compressors and blowers should also always be reevaluated and can be used to assist heaters, boilers, dryers, or even site instrument air systems if there are no suitable process uses.

Pumps

Pumps are relatively inexpensive compared with compressors and will often be replaced entirely in a revamp. Pumps are discussed in more detail in Section 20.7.

The most commonly used pumps are centrifugal pumps. Centrifugal pumps exhibit pressure-flow behavior similar to that shown for centrifugal compressors in Fig. 2.20. A typical pump curve is given in Fig. 20.15. The same pump can deliver a set of different performance curves depending on the impeller diameter and motor speed. The designer can therefore sometimes obtain the required performance just by selecting new impellers for the existing pumps.

Control valves

All control valves must always be rated to confirm that they are correctly sized for the revamped design case and will give the desired controllability, turndown, and ability to meet different operating scenarios. Control valve design and sizing are discussed in Section 20.11.

When parallel equipment is introduced into the revamped flowsheet, the design engineers must determine how the flow is to be split between the existing and new equipment. A simple T-junction or branch will often not be effective, as the rate of fouling or pressure-drop accumulation of the new and old equipment will not be the same, even if they have the same capacity and design, and flow would then preferentially go through one piece of equipment, leading to poor performance of both. The desired split ratio is rarely 1:1 and may need to be adjusted once the plant is put into operation. The most common approach is to put a new control valve on the branch that leads to the new equipment. The existing equipment is likely to be limited by pressure drop, so the new equipment can be designed with lower pressure drop to allow for the pressure drop associated with the control valve. A less costly, but less effective, alternative is to use a manual valve or even a restriction orifice in the bypass line and make manual adjustments until the desired flow ratio is obtained.

Control valves that act discontinuously and handle low flow rates will sometimes be suitable for reuse in the revamped plant as long as the new actuation rate is acceptable. Control valves on main plant flows will usually need to be replaced, as will control valves on gas or vapor streams. The revamped flowsheet does not need to indicate which existing valves are reused and which are replaced, but should show all new control valves that are added.

2.6 Synthesis of novel flowsheets

The terms *process synthesis* and *conceptual process design* are used for the invention of completely new process flowsheets. As stated previously, very few entirely new designs are developed commercially because of the high financial risks inherent in using unproven technology. The primary goals of process synthesis are therefore to reduce commercialization risk and to maximize economic attractiveness so as to generate sufficient financial reward to balance the risk.

Process synthesis has been the subject of a great deal of academic and industrial research over the past 40 years. Many problems that were previously solved using inspired guesswork can now be formally posed and optimized. The use of process simulation programs has also made it much easier to evaluate and optimize alternative flowsheets; see [Chapter 4](#) for more on process simulation. Several prominent researchers in the field of process synthesis have written textbooks on process design that strongly emphasize process synthesis. These are listed in the bibliography at the end of this chapter. Several excellent books have been written on process synthesis in its own right: Rudd et al. (1973); Douglas (1988), and El-Halwagi (2006), as well as on aspects of process synthesis, such as distillation sequencing (Doherty and Malone, 2001), mass integration (El-Halwagi, 2017), and heat integration (Kemp, 2007; Shenoy, 1995). Although it is beyond the scope of this book to cover all aspects of process synthesis, this section sets out an overall framework for flowsheet synthesis that addresses the key issues encountered in developing and commercializing new processes. The reader is encouraged to read the books listed here and in the bibliography for more insights into the subject.

2.6.1 Overall procedure for flowsheet synthesis

Most efforts to systematize process synthesis begin by setting out a sequence or hierarchy of steps for the designer to follow. Design hierarchies recognize that some steps need to come before others and should guide the designer to eliminate unattractive options and focus effort on designs that are most likely to be successful.

The most intuitively obvious design hierarchy is the so-called *onion diagram*. [Fig. 2.24](#) shows a version of the onion diagram given by [Smith \(2016\)](#). The onion diagram represents a design starting with the reactors; adding separation and recycle systems; then proceeding to add heat recovery, utility systems, and environmental systems. [Rudd et al. \(1973\)](#) proposed a more theoretically abstract synthesis hierarchy that essentially follows the same steps but included an additional step of integrating reaction, mixing, separation, or change of state tasks into unit processes or operations.

[Douglas \(1988\)](#) set out a somewhat different approach, summarized in [Table 2.1](#). Douglas emphasized early introduction of process economics to guide the elimination of weaker alternatives, with continuous refinement of the economic model as more detail is added to the flowsheet. This is a useful approach, as it can highlight deficiencies in the design at an early stage.

The reality of process development in industry is usually a lot less systematic than the idealized picture painted by academic researchers. In industrial practice, process development is more interdisciplinary, more iterative, and much less linear than the simple synthesis models suggest. The flowsheet synthesis step is usually part of a larger

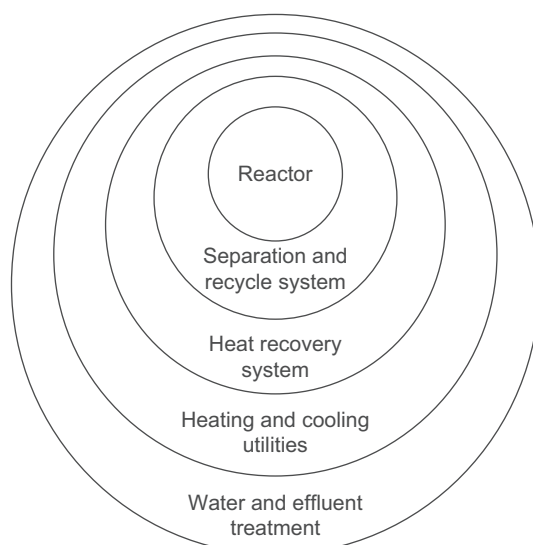


FIG. 2.24 Onion diagram. From Smith, R. [2016]. Chemical process design and integration [2nd ed.] Wiley.

TABLE 2.1 Hierarchy of process synthesis decisions

1. Batch vs. continuous
2. Input–output structure of the flowsheet
3. Recycle structure of the flowsheet
4. General structure of the separation system
 - a. Vapor recovery system
 - b. Liquid recovery system
5. Heat exchanger network

Data from Douglas, J. M. (1988). Conceptual design of chemical processes. McGraw-Hill.

effort that involves chemists, biologists, and other engineers and includes laboratory and pilot plant experiments to determine reactor performance and establish yields and product recoveries. The engineers working on synthesis seldom have all the data that are needed to properly optimize the design and often must guide the research members of the team to collect additional data under conditions that will be more favorable to process economics. Many industrial processes involve multiple reaction steps carried out in sequence with intermediate separation steps, and it may be difficult to assess the performance of later steps without good information on the nature and quantity of by-products that are carried over from earlier steps in the sequence. The design team therefore needs to form a rough impression of the PFD and economics, using minimal information on process chemistry, so as to develop an understanding that can guide the efforts of the research team.

Fig. 2.25 sets out an approach to flowsheet synthesis that sets flowsheet development in the context of working with a research team to establish yields and reactor performance. The Douglas hierarchy and onion diagram form substeps in this procedure, as described next.

Step 1. Initial economics

The very first step should be to collect prices for feeds and products and confirm that the cost of production will be attractive if a stoichiometric yield is obtained. If the cost of feed is more than the product value, then there is no hope of developing an economically attractive process and the work should be stopped unless the team has strong evidence that prices will change in the future. This step is particularly important when assessing nontraditional feeds; for example, when looking at processes for converting food-based renewable feeds into chemicals.

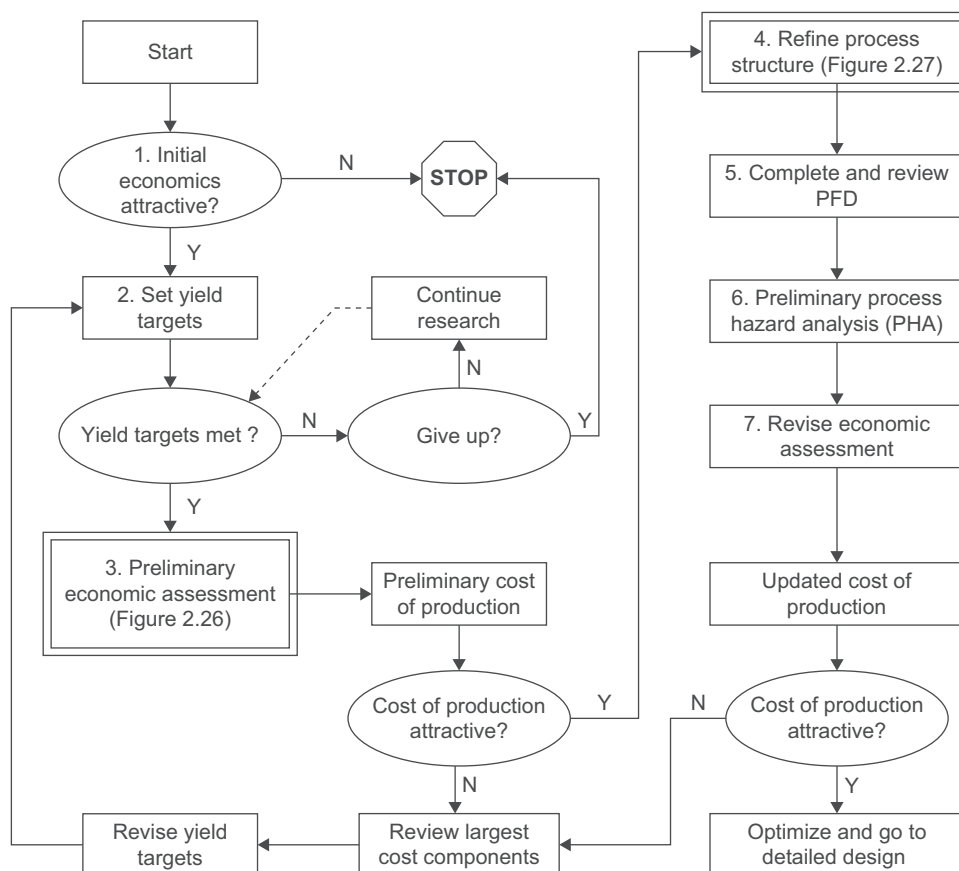


FIG. 2.25 Procedure for process synthesis.

Step 2. Set yield targets

The research team needs to set realistic targets that will lead to an attractive process. The term *yield targets* includes targets for by-product selectivity as well as main product selectivity and conversion. Methods for setting and revising yield targets are given in Section 2.6.3. The researchers will generally need to carry out process development experiments to optimize reactor conditions and catalyst, enzyme, or organism performance to meet the yield targets.

On the first pass through the procedure, the designers may choose to just accept whatever yields and selectivities the research chemists or biologists have already established. When more information on process economics has been generated, the targets can be revised and improved.

If the yield targets are not met, the company must make a strategic decision on whether to continue or abandon the research. Research discoveries are often serendipitous and can be hard to plan. Companies often choose to allow a low level of research activity to continue over a long period once clear success criteria have been established.

Step 3. Preliminary economic assessment

The goal of a preliminary economic assessment is to arrive at a preliminary estimate of the cost of production once the yield targets have been met. The substeps in carrying out a preliminary economic assessment are illustrated in Fig. 2.26. This procedure is similar to that of Douglas (1988), but it should be emphasized that less detail is put into the design and the goal is not to arrive at a PFD or even a detailed block flow diagram at this point.

The components of cost of production and methods for calculating each component are discussed in detail in Chapter 8. For most processes, 80% or more of the cost of production will be feedstock cost minus credits for any economically viable by-products. The rest of the cost of production is chiefly made up of utility costs (mostly energy), fixed costs (mostly labor), and annualized payments to generate an expected return on the capital investment. The split between these depends on the type and scale of the process. Small-scale batch processes will have a higher proportion of fixed costs, whereas large-scale petrochemicals or solids-handling plants will have a higher proportion

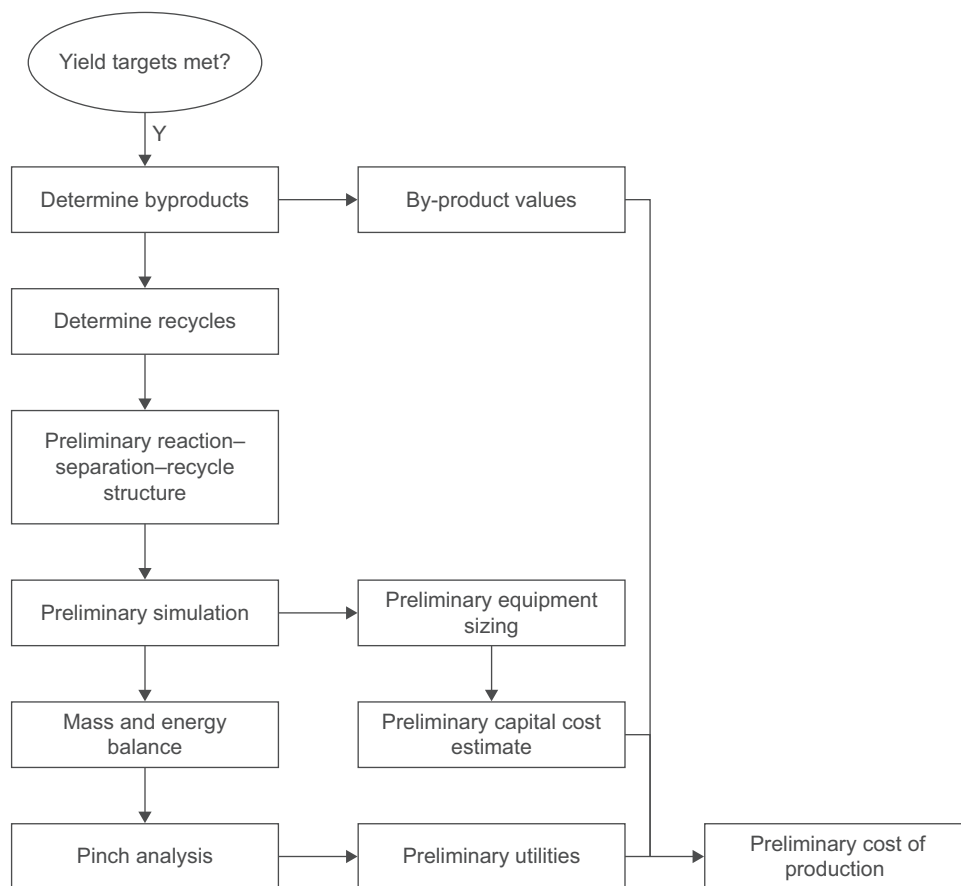


FIG. 2.26 Preliminary economic assessment.

of utility costs. In a preliminary economic assessment, the designer seeks to make a quick estimate of these major components of cost, and hence determine whether the process will be able to make product at an attractive price.

The first step is to look at the yields and identify any significant by-products. By-products can be formed by the reaction stoichiometry, by side reactions, or from extraneous components in the feeds. By-products must be refined and sold, treated as waste streams, or recycled to extinction in the process. A quick examination of the yields and prices of the by-products is usually sufficient to establish which are suitable for recovery. The economic assessment of by-product recovery is discussed in more detail in Section 8.2.3.

Once the designer has a notion of which by-products are worth recovering and which by-products must be recycled, a preliminary reaction-separation-recycle structure can be sketched. It is not important to have the best or optimal flowsheet at this point, and the design team may want to propose a few alternatives to see which is least costly. A process simulation model can then be built and used to generate a mass and energy balance and obtain rough sizing of the major process equipment. The preliminary simulation should include all recycles, reactors, and separation equipment and should capture all changes in temperature and pressure. It does not need to include a heat recovery design and should use heaters and coolers instead of heat exchangers whenever there is a change in temperature. The use of commercial programs for process simulation is discussed in Chapter 4. The preliminary simulation model does not have to be built using a commercial simulation program, but it will usually be convenient to do so, so that the model can be extended as the flowsheet is subsequently refined.

The preliminary equipment sizes from the simulation model (or hand calculations) can be used to develop an initial estimate of the capital cost of building a plant. Estimation of capital costs is covered in Chapter 7. The capital cost is then annualized so that it can be added to the other costs of production. Annualization of capital costs is covered in Section 9.7. In a preliminary economic assessment, the designers usually annualize capital costs by dividing by 3. The basis for this rule of thumb is set out in Section 9.7.2.

The process energy consumption and utility costs are difficult to assess without completing a process simulation and energy balance. Most processes have significant opportunity to reduce energy costs by heat recovery, so simply

adding up all the heating and cooling duties would be a gross overestimate. Instead, a first estimate of energy consumption can be made by carrying out the targeting step of pinch analysis to get hot and cold utility targets. Pinch analysis and other heat recovery methods are described in [Chapter 3](#). At this point in the design it is not necessary to design the heat recovery system, as the targets are adequate for the preliminary economic assessment.

The preliminary estimates of main product and by-product production rates, feed and energy consumption, and capital cost can be used to make a preliminary estimate of the cost of production, as described in [Chapter 8](#). If the cost of production appears attractive, the design team proceeds to the next step. If not, the economic assessment can be used to highlight the major components of cost that must be reduced to make the process economically interesting. Having identified the cost components that must be addressed, the design team can either look at alternative flowsheets that reduce these costs or else set more aggressive yield targets and go back to the research stage.

Step 4. Refine process structure

If the preliminary economic assessment indicates that the process is potentially economically attractive, then it is important to develop a complete PFD and make sure that no costs have been overlooked. The steps in completing a more rigorous design are shown in [Fig. 2.27](#). It can be seen that these follow roughly the same sequence as the onion diagram of [Fig. 2.24](#).

The first step is to optimize the reaction–separation–recycle structure of the flowsheet and confirm the yields under the optimal conditions. The preferred conditions can be estimated by optimization of the preliminary simulation model and economic model. Additional experimental data may be needed if the optimal conditions are different from the conditions originally proposed. The reactor designs must be tested and yields confirmed in the presence of recycle streams, which may require construction of a pilot plant that can operate in recycle mode. The design of reactors is described in [Chapter 15](#).

The design of the separation systems encompasses not only those separations associated with product recovery and recycles but also feed purification, product purification, and by-product recovery. The design of separation processes is covered in detail in [Chapters 16, 17, and 18](#). In some cases, product purification or by-product recovery will require additional reaction steps. For example, in the recovery of ethylene produced by steam cracking of light hydrocarbons, it is easier to hydrogenate by-product acetylene than to separate it by distillation ([Fig. 2.28](#)).

When the by-product separation and recovery sections have been designed in more detail, the attractiveness of recovering the by-products can be revisited. If the costs of producing the by-products are excessive, the designer should revisit the reaction–separation–recycle structure or return to the preliminary economic analysis step.

When designing the reaction and separation steps, as much use as possible should be made of proven process subsections. If a particular reaction, separation, recovery, or purification step is already practiced commercially, then the same method will probably be least costly and will most likely have the least technical risk for the new design. Borrowing proven concepts from established technology is one of the most effective strategies for reducing commercialization risk. Some caution is needed when applying this strategy if the new case is not identical to the commercially proven design, and the designer should take particular care to ensure that slight differences do not lead to potential safety or operability issues. The designer must also ensure that there are no active patents on the features that are borrowed so that there is freedom to practice them.

Once the major process equipment has been specified, the design team should have a good idea of the stream temperature and pressure requirements, and the heat recovery system can be designed. Process heat recovery is described in [Chapter 3](#), and heat transfer equipment is covered in [Chapter 19](#).

The designs of the plant hydraulics and control system are interlinked, as control valves introduce additional pressure drop into the process and can create a requirement for additional pumps. Once the major equipment, including heat exchangers, has been specified, a preliminary PFD can be drafted, which can be developed into a full PFD by adding the location of control valves, pumps, and compressors. The design of plant control systems and location of control valves is discussed in [Chapter 5](#). The design of hydraulic equipment is covered in [Chapter 20](#), and solids handling systems are described in [Chapter 18](#). When all the equipment has been added to the PFD, the process simulation can be updated to produce mass and energy balances to complete the flowsheet. The PFD is then ready for review.

Step 5. PFD review

Review of a PFD is one of the most important steps in flowsheet development. A full PFD review is always carried out in design, regardless of whether the process is a revamp or new unit, or whether it uses novel or proven technology. This vital step is discussed in more detail in [Section 2.7](#).

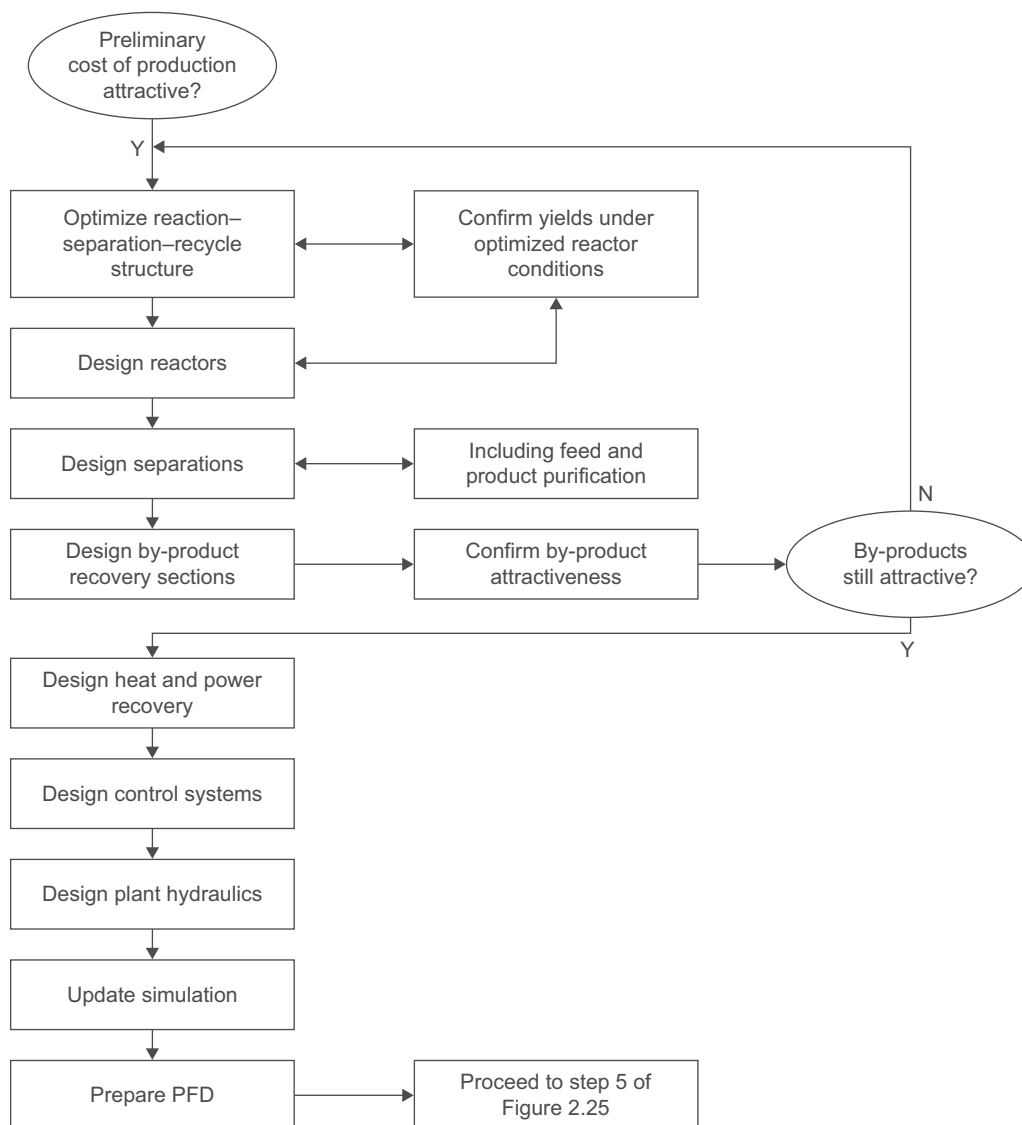


FIG. 2.27 Refining the process flowsheet.

Step 6. Preliminary process hazard analysis

When a completed PFD and mass and energy balance are available, a preliminary process hazard analysis (PHA) can be carried out. A PHA will identify major hazards inherent in the process and may indicate a need to alter some process conditions, substitute different equipment, or completely redesign sections of the process. If the preliminary PHA identifies major modifications to the PFD, the design team should go back to the relevant stage of the procedure and generate a safer alternative design. The role of safety in design is discussed in detail in [Chapter 10](#).

Step 7. Revise economic assessment

The completed PFD and mass and energy balance allow the design team to make more accurate designs of the process equipment, and hence arrive at more accurate estimates of the capital cost and cost of production. If the process still appears to be attractive, then it may be worth developing as an investment, and other methods of economic assessment will be used to determine a viable project for implementing the technology; see [Chapter 9](#). The models that have been developed now have sufficient accuracy to enable more rigorous optimization and can be used as a starting point for detailed design of the plant and equipment. Optimization methods in process design are discussed in [Chapter 12](#).

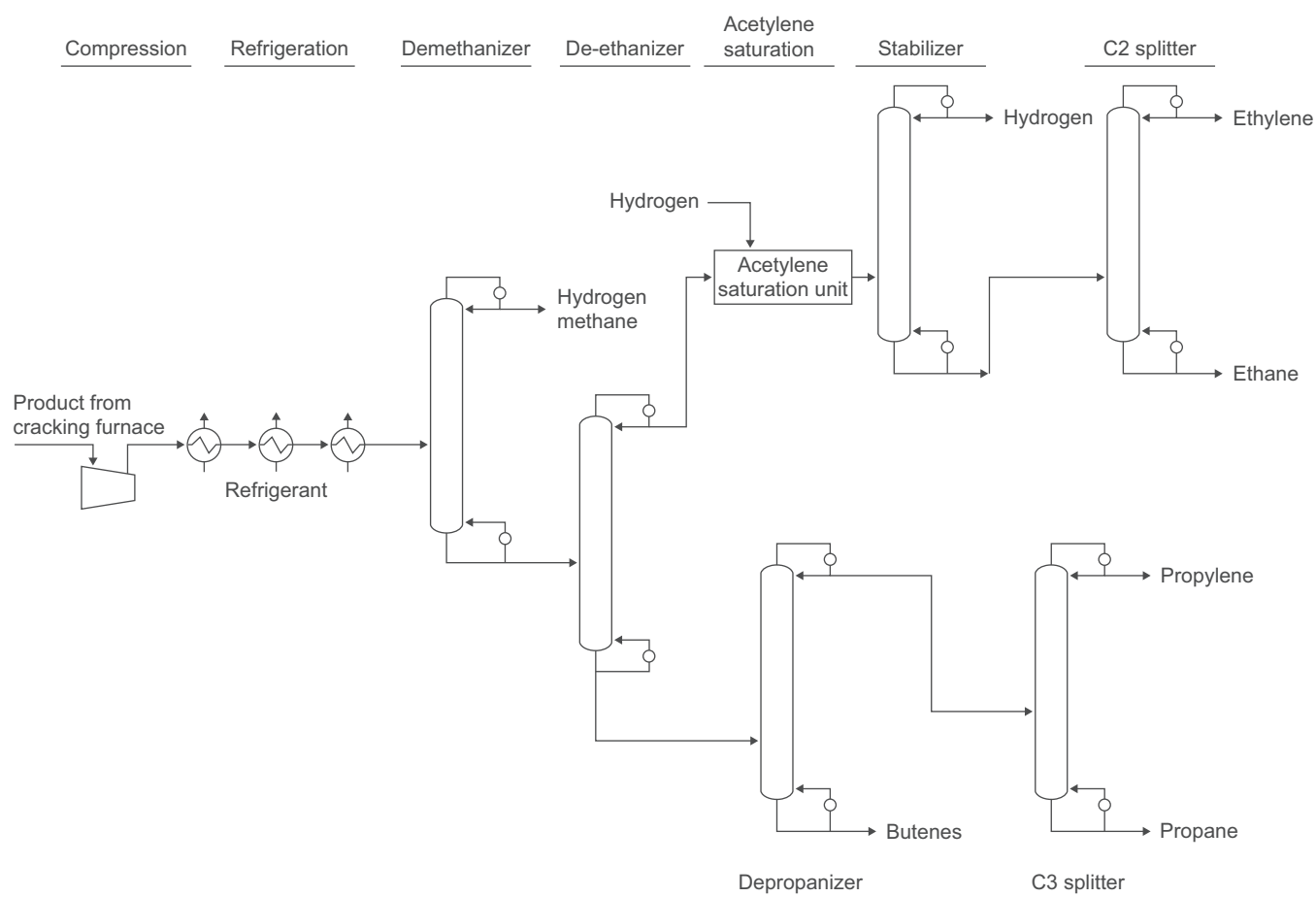


FIG. 2.28 Ethylene recovery from steam cracking.

If the updated cost of production is no longer attractive, the added detail can be used to further define the major cost components and identify areas for process improvement. This will often involve reducing by-product and recycle handling, and hence translate into setting tougher yield targets.

2.6.2 Economic analysis in process synthesis

In process synthesis it is important to use economic analysis to inform decision-making at every step. The procedure outlined in Fig. 2.25 begins with a rough economic analysis and then adds detail to this analysis as information is accumulated. Like an artist beginning with a rough pencil sketch and then filling in details and adding colors, the process design engineer needs to have an overall sense of the composition before getting into the details.

The procedure set out in Fig. 2.25 has three economic check points at steps 1, 3, and 7, corresponding to the initial, preliminary, and updated estimates of the cost of production. At these check points the estimated cost of production should be compared with the product sales price that the marketing organization has forecasted. Usually, the criterion for success will be that the cost of production must be low enough to ensure an acceptable return on the capital deployed. Forecasting of prices and calculation of cost of production are covered in Chapter 8, and methods of economic analysis are described in Chapter 9. Although these three steps in the procedure are formal checks, experienced designers do not wait until the check point to calculate process costs. As soon as information is developed, its impact on the cost of production should be determined. In general, costs accumulate as detail is added to the design, so the design team wants to be aware of large costs as early as possible so that they can start considering alternative design features.

Companies usually use other economic analysis methods, such as net present value (NPV) and internal rate of return (IRR), instead of cost of production when assessing capital investment projects; however, very few companies launch a capital project without having already completed the process synthesis. At the synthesis stage, cost of production is the most useful economic measure, as it is very easily factored into components such as raw material

costs, by-product values, energy costs, etc. Understanding the components of cost of production can help the design team focus on areas of high cost if the target cost of production is not achieved.

2.6.3 Use of targets in process synthesis

Design engineers use targets as a means of setting bounds on design performance that can quickly eliminate unattractive options. Targets also help designers and researchers to focus their efforts on areas that will most effectively improve economic performance. Several different types of targets are used in the synthesis procedure set out in Figs. 2.25, 2.26, and 2.27:

- The cost of production is compared with a price target set by the marketing organization at steps 1, 3, and 7.
- The design team sets targets for the research team at step 2.
- Hot and cold utility targets calculated by pinch analysis are used for initial estimates of process energy consumption.
- The preliminary economic analysis sets targets for capital cost and the components of cost of production that the design team must confirm as they refine the process structure and fill in the PFD.

The basis for targets should always be clearly stated. Whenever possible, targets should be calculated from economic criteria, and the assumptions in the calculations should be made explicit. For example, a badly stated yield target would be:

“Find me a catalyst that doesn’t make by-product X.”

A better way of stating this target might be:

“If reactor selectivity to by-product X is $<0.5\%$ of selectivity to the main product, we can eliminate a separation and purification section of the process, with expected 15% savings in capital cost and 20% savings in energy cost.”

Targets must not be unrealistically tough, or they will never be achieved and will not be taken seriously. Lenient yield targets are usually not as problematic, as they tend to lead to failure at the preliminary economic assessment stage and are then revised to something more realistic. Lenient targets for cost of production are dangerous, as they allow the design to go forward, and much time and effort can be wasted before harsh economic reality kills the project. Good price forecasting and market analysis are critically important in setting the cost of production targets. These topics are discussed in Chapter 8.

It is important to understand whether a target should be treated as a hard constraint or a soft constraint. Companies sometimes address this question by providing *must-have* targets that are hard constraints and *should-have* targets that are soft constraints. The design team can then reject designs that do not meet the hard constraints, but keep concepts that come close to the soft targets. Soft and hard targets are explored in Example 2.8.

Example 2.8

The marketing group is planning to launch a new product and has forecasted that the mean price for the product will be 5 \$/kg, normally distributed with standard deviation 40 ¢/kg. The cost of the stoichiometric amount of feed required to make the product is 3 \$/kg. Propose preliminary targets for cost of production and yields.

Solution

If the forecast is accurate and the mean product price is 5 \$/kg, then there is a 50% probability the project will have economic success if the cost of production (including capital recovery) is 5 \$/kg. Using the standard deviation given in the forecast, we can form the following table:

Cost of production (COP), \$/kg	Probability of success
3.80 ($= 5 - (3 \times 0.4)$)	99.9%
4.20 ($= 5 - (2 \times 0.4)$)	97.7%
4.60 ($= 5 - (1 \times 0.4)$)	84%
5.00	50%
5.40 ($= 5 + (1 \times 0.4)$)	16%

We can note immediately that this process would pass the initial economic assessment even if we chose a target cost of production with 99.9% chance of success.

The probability of success that we require depends on how risk averse or aggressive the company is. A 98% chance of success might be too conservative and would give a high chance that the project would not meet the targets. A 50% chance of success would probably be too aggressive and would allow the project to go forward and spend money with low likelihood of financial success. As a compromise, management might set a *must-have* target for COP of 4.60 \$/kg, with a *should-have* target of 4.40 \$/kg. Note that we are not constrained to using integral multiples of the standard deviation in setting the targets; for example, a price target of 4.34 \$/kg ($= 5 - (1.65 \times 0.4)$) corresponds to 95% probability of success and would be equally easy to justify as a *must-have* or *should-have* target.

The COP target can now be translated into preliminary yield targets. In Section 2.6.1 it was stated that feed costs are typically at least 80% of COP. Using this rule of thumb, we can state:

$$\text{Target feed cost} = 0.8 \times \text{target COP}$$

The yield targets obviously depend on the number of feeds, the relative costs of individual feeds, and the number of reaction and product recovery steps. From Equations 2.4 and 2.6 we know:

$$\text{Yield} = \frac{\text{product formed}}{\text{stoichiometric product formed}}$$

so:

$$\text{Target plant yield} = \frac{\text{stoichiometric cost /kg}}{\text{target feed cost /kg}}$$

So if the target COP is 4.60 \$/kg:

$$\text{Target plant yield} = \frac{3}{0.8 \times 4.60} = 0.815$$

Note that this is the target yield over all the steps in the process. If we assume that we lose roughly 5% of the product during all the steps of product recovery and purification, then:

$$\text{Target plant yield from reactors} = \frac{0.815}{0.95} = 0.858$$

If we have two reaction steps and one feed is more expensive than the others, we could further decompose the yield target into targets for each step:

$$\text{Target plant yield from reactors} = Y_1 Y_2$$

where Y_1 and Y_2 are yields of reaction steps 1 and 2. We could set equal targets for each step:

$$Y_1 = Y_2 = \sqrt{(0.858)} = 0.926$$

or we could use our knowledge of process chemistry or biology to define a suitable allocation between the steps.

Note that the targets calculated are plant yields, not reactor yields. If costly unconverted feeds can be recycled, the plant yields translate into reactor selectivity targets, not reactor yield targets (see Section 2.3.3).

Note also that even in this simplistic example, the apparently very favorable economics quickly translated into rather tough targets for reactor performance.

2.6.4 Use of heuristic rules in process synthesis

Heuristic is an adjective meaning “of, or pertaining to, or based on, experimentation, evaluation or trial and error methods,” which pretty much sums up most engineering knowledge. The terms *heuristic rules* and *design heuristics* are commonly used to describe rules of thumb and design guidelines that have been developed based on experience. Experience is good, but it is rarely acquired instantly or bought cheaply. Design guidelines based on generalizations are only useful if the designer has sufficient experience to understand when to apply the guideline and when to make an exception.

There is often confusion about what constitutes a design heuristic. Consider the following statements, most of which have been made in this chapter:

1. *"The heat of condensation of steam is about 2000 kJ/kg."*
2. *"Pressure drop is usually proportional to velocity squared."*
3. *"Feedstock costs are typically at least 80% of the total cost of production."*
4. *"Capital costs can be annualized by dividing by 3."*
5. *"When designing reaction and separation steps, use proven process subsections as much as possible."*

Statement 1 is a convenient data approximation. It is accurate within $\pm 10\%$ for saturated steam over a temperature range from 100 °C to 240 °C, which covers most temperatures at which steam is used for process heating. Remembering this fact may save an engineer some time when carrying out hand calculations, but it does not provide any guidance for design.

Statement 2 is a convenient summary of several correlations for pressure drop. It may be very useful for making quick calculations in revamp designs, but again provides no guidance for design.

Statement 3 is a generalization that can be useful as a rough check on cost of production calculations. As illustrated in Example 2.8, it can also be used as a basis for setting initial targets in process synthesis. It is, however, rather too general to provide guidance in design.

Statement 4 is one way of annualizing capital costs. It would be equally valid to state *"Capital costs can be annualized by dividing by 2"* or *"...by dividing by 5,"* depending on the assumptions made. The basis for deriving these numbers is given in Section 9.7.2.

Statement 5 is clearly a design guideline based on a general desire to minimize the number of unproven concepts in the flowsheet.

Although all of these statements are useful as rules of thumb and can help make quick calculations to assess a design, only statement 5 actually provides guidance on how to design a process. It might be useful for an engineer to recall approximate data and generalizations, and several compilations of such rules of thumb have been written (Fisher, 1991; Hall, 2017; Hicks & Chopey, 2012), but these are only helpful in process design when making quick calculations in meetings.

Some design texts provide extensive lists of rules of thumb and selection guidelines. That approach is not adopted here, as heuristics easily lose their meaning when taken out of context. For example, one design text gives the rule for vessel design: *"optimal length to diameter ratio = 3."* This is questionable even for horizontal and vertical flash drums (see Section 16.3), but is clearly nonsense when wrongly applied to reactors and distillation columns. Inexperienced engineers often have difficulty determining when to apply such heuristic rules, so in this text all shortcut calculations, convenient approximations, and design guidelines will be presented and explained under the relevant design topic.

The most important heuristic rule, which should always be followed, is:

"Never use a heuristic rule unless you understand where it came from and how it was derived."

2.6.5 Role of optimization in process synthesis

Optimization methods are used in process synthesis to select the best flowsheet options, process conditions, and equipment sizes. The designer must be reasonably sure that design alternatives have been optimized before selecting between them. Optimization underpins all design decisions, and the subject is addressed in more detail in Chapter 12.

The problem that is often encountered when applying optimization methods in process synthesis is that insufficient data are available to properly formulate an optimization problem. For example, it would be good to optimize reactor performance as soon as possible, but the research team may not yet have collected data in the presence of recycle streams or at conditions close to the optimal reactor conditions. Under the circumstances, the designer must optimize the design with the data available and then revisit the optimization later when more data have been collected and the reaction kinetics model has been updated.

Most processes are too complex to formulate a single optimization problem that includes all possible structural flowsheet variations, as well as all continuous process variables. Instead, different aspects of flowsheet synthesis are usually treated as separate optimization tasks. It is important, however, to have an overall optimization model that

captures the major design trade-offs. The cost of production model developed in the preliminary economic assessment can serve as an initial model for optimization. The optimization of subproblems is discussed in Section 12.5.

2.7 PFD review

The most important step in developing a process flowsheet is for the PFD to be rigorously reviewed. This is true regardless of whether the flowsheet is for an established design, a revamp, or a newly invented process. The purpose of a PFD review is to review the design decisions and assumptions and ensure that the flowsheet is complete and shows *all* the equipment needed to really operate the process.

A PFD review is usually attended by the design team and a few outside experts. These may include:

- Senior managers
- Technical experts on process design
- Technical experts on process chemistry, catalysis, or biology
- Equipment or plant design experts
- Process safety experts
- Metallurgists
- Plant operations staff
- Plant mechanical engineers
- Process control engineers

Some companies have rules and procedures governing who must be present at a PFD review, but the review can be effective with only a few people as long as they engage actively with the team.

2.7.1 PFD review procedure

A PFD review is usually carried out as a group exercise. A large printout of the PFD is typically taped or pinned to a wall so that the group can mark up corrections, notes, and other revisions as the review proceeds. The PFD is usually drawn well spaced out to leave room for additions, and may run to several sheets of drawings. One member of the group takes notes and documents any actions that are agreed to or concerns that are raised. If the heat and material balances and stream conditions are not shown on the PFD, printouts are usually provided for the reviewers.

For complex processes, a PFD review may begin with a brief overview of the process chemistry and block flow diagram to establish the context for the reviewers. In some cases the design basis assumptions are also reviewed at the start.

The main part of a PFD review is a “walk through” of the process by the process design engineer. Starting with one feed stream, the designer follows the stream from storage through all the process operations that it encounters. At each process operation, the designer explains the purpose of the operation, the design criteria, and the resulting condition of the stream at the exit. For example, the feed section shown in [Fig. 2.29](#) would be described as follows:

Feed of 99% pure technical grade A leaves floating roof storage tank T101 through line 101 at ambient conditions. The governing ambient temperature for heater design is winter low temperature of -5°C and for pump design is summer high temperature of 30°C . Stream 101 is pumped by centrifugal pump 101 to a pressure of 10 bar gauge, forming stream 102. Pump 101 has a standby spare, shown. Flow of feed A is regulated by flow control valve FCV100, with design pressure drop 1.3 bar. Stream 103 exits FCV100 and is sent to steam heater E101. The purpose of heater E101 is to heat feed to the reaction temperature of 180°C . High-pressure steam at 240°C is used as heat source. The steam rate is controlled by temperature controller TC101, which receives input from temperature indicator on the process stream, leaving E101 in Stream 104. The steam has been placed tube side in E101 because the process stream is nonfouling and steam is at high pressure, so this is expected to lead to the lowest-cost design. A pressure drop allowance of 0.7 bar has been assigned to E101. Stream 104 leaves E101 at the desired reactor feed temperature of 180°C and at reactor pressure 8 bar gauge and enters reactor R1.

As the designer steps through the PFD, the review group asks questions to challenge the design assumptions and identify potential missing equipment. In the previous example, some relevant questions could have been:

- Is it necessary to pass the feed through a filter before FCV100 to remove any crud that accumulated in the tank or came in with the feed?
- Why use high-pressure steam to heat all the way from ambient temperature? Couldn't process heat recovery be used for at least part of the heating?

TABLE 2.2 Sample questions that can be asked in a PFD review

Process Section	Questions
Feed preparation	How is (each) feed delivered?
	How is (each) feed stored?
	How much inventory of (each) feed is required?
	How is feed transferred from storage to the process?
	How is the rate of feed supply controlled?
	Is any feed pretreatment necessary before the feed is sent to the process?
	For solid feeds, is any feed size adjustment necessary?
	Is any heating or cooling needed before the feed is sent to the process?
Reaction	What are the reacting species?
	What side reactions occur?
	What are the reaction conditions and why were they chosen?
	How are the reaction conditions maintained or controlled?
	How are inventories of solids, liquid, or vapor in the reactor controlled?
	What are the reactor design specifications (e.g., residence time, interfacial area)?
	What are the estimated reactor yield and selectivity?
	What reactor type was chosen and why was it selected?
	Is a catalyst used? If so, is the catalyst stable or does it require periodic regeneration?
	Is heat addition or removal necessary?
Product Recovery	Are there specific safety issues that should be considered?
	(Additional information on reactor design is given in Chapter 15 , which may prompt more questions)
	What is the purpose of each separation?
	What are the process conditions (temperature, pressure, etc.)?
	What are the equipment specifications (recovery, purity, etc.)?
	Why was a particular separation selected?
	Is heat removal or addition necessary?
	Can heat addition or removal be accomplished by process-to-process heat transfer?
	How are inventories of vapor, liquid, or solid controlled in each operation?
	How is the operation controlled to achieve the desired specifications?
Purification	Are there specific safety issues that should be considered?
	What are the final product specifications?
	What are the specifications for any by-products?
	What are the specifications for any effluents discharged to the environment?
	What are the specifications for any recycle streams returned to the process?
	What are the specifications for any waste streams sent to disposal?
	How are final purity specifications on any stream leaving the process achieved?
	How is the process controlled to ensure that purity specifications can be achieved?

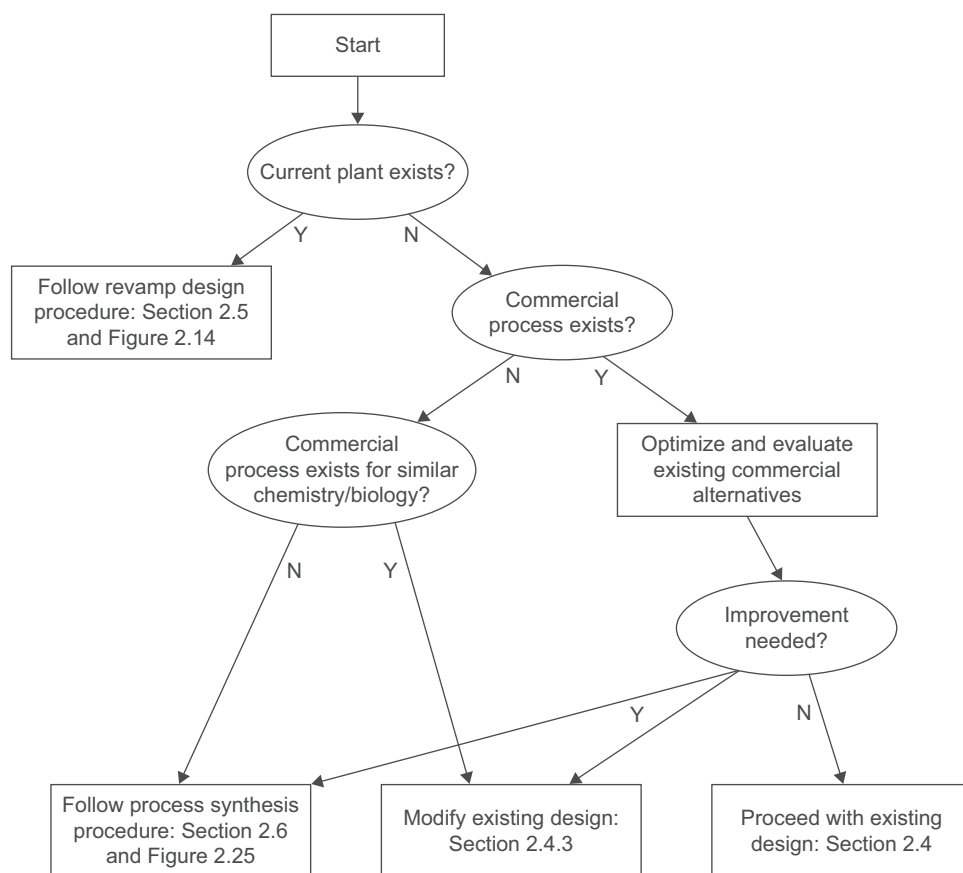


FIG. 2.30 Overall procedure for flowsheet development.

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2.10 Nomenclature

		Dimensions in MLT
A	Area for heat exchange	L^2
$C_{p,f}$	Specific heat capacity of feed	$L^2 T^{-2} \theta^{-1}$
$C_{p,i}$	Specific heat capacity of stream i	$L^2 T^{-2} \theta^{-1}$
$C_{p,p}$	Specific heat capacity of product	$L^2 T^{-2} \theta^{-1}$
F	Unknown flow, Example 2.6	MT^{-1}
F_{cw}	Flow rate of cooling water	MT^{-1}
F_t	Log mean temperature difference correction factor (see Chapter 19)	—
m_f	Mass flow rate of feed	MT^{-1}
m_i	Mass flow rate of stream i	MT^{-1}
m_p	Mass flow rate of product	MT^{-1}
Q	Rate of heat transfer	$ML^2 T^{-3}$
SV	Space velocity	T^{-1}
T_1	Unknown temperature, Example 2.7	θ
T_2	Unknown temperature, Example 2.7	θ
T_3	Unknown temperature, Example 2.7	θ
T_s	Steam temperature	θ
T_w	Cooling water outlet temperature	θ
ΔT_i	Change in temperature of stream i for a stream undergoing sensible heat changes	θ
ΔT_{lm}	Logarithmic mean temperature difference	θ
ΔT_m	Mean temperature difference for heat transfer	θ
U	Overall heat transfer coefficient	$MT^{-3} \theta^{-1}$
V	Reactor volume	L^3
V_{cat}	Catalyst fixed bed volume	L^3
v	Volumetric flow rate	$L^3 T^{-1}$
X	Unknown flow, Examples 2.1, 2.5	MT^{-1}
Y	Unknown flow, Example 2.5	MT^{-1}
Z	Unknown flow, Example 2.5	MT^{-1}
τ	Residence time	T

Common PFD abbreviations

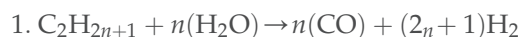
BFW Boiler feed water
CW Cooling water
CTW Cooling tower water
DBO Design by others (used to refer to a section of a flowsheet that is purchased from a vendor as a packaged plant)
DIW Deionized water
HPS High-pressure steam
LPS Low-pressure steam
MPS Medium-pressure steam
NNF Normally no flow (used to indicate discontinuous operations in a continuous plant flowsheet)
ROW Reverse-osmosis water

Abbreviations for designating process control instruments are given in [Chapter 5](#).

2.11 Problems

- 2.1. Cinnamic aldehyde (a fragrance compound) can be made by base-catalyzed aldol condensation reaction between benzaldehyde and acetaldehyde. The feeds are contacted with sodium hydroxide in a stirred tank reactor. The product is neutralized and washed with water to remove salts. The washed product is usually separated by batch distillation, in which unreacted feeds are recovered first, followed by product, and a polymeric waste is left as residue. The batch distillation product can be further purified by vacuum distillation. Sketch a block flow diagram of the process.
- 2.2. Cumene is made by alkylation of benzene with propylene over zeolitic catalyst. To maximize selectivity to desired products, several beds of catalyst are used inside the same reactor. A mixture of feed and recycle benzene enters the top of the reactor and the feed propylene is split so that a portion of the propylene is fed to each catalyst bed. A large excess of benzene is used to minimize propylene oligomerization and ensure complete reaction of propylene. The reactor product is cooled and sent to a stabilizer column that removes any light hydrocarbons. The bottoms from this column are sent to a benzene column that recovers benzene overhead for recycle to the alkylation and transalkylation reactors. The bottoms from the benzene column are distilled to give cumene product and a heavy stream. The heavy stream is further distilled in a heavies column to give a mixture of dipropyl- and tripropyl-benzene overhead and a heavy waste stream as bottoms. The distillate from the heavies column is sent to a transalkylation reactor and reacted with excess benzene. The product from the transalkylation reactor is returned to the benzene column. Sketch a block flow diagram of the process.
- 2.3. Cyclosporin A is produced by fermentation using either *Cylindrocarpum lucidum* Booth or *Tolypocladium inflatum* Gams. The fermentation is carried out in batch reactors, which are filled with a feed medium, inoculated with the fungi, and aerated for a period of 13 days. The reactor product is milled and extracted with 90% methanol. The methanol is evaporated off to give an aqueous solution that is then extracted with ethylene chloride. The organic solution is evaporated to dryness, and then the product is purified by chromatography in methanol over aluminum oxide or silica gel. Sketch a block flow diagram of the process.
- 2.4. The composition of a gas derived by the gasification of coal is volume percentage: carbon dioxide 4, carbon monoxide 16, hydrogen 50, methane 15, ethane 3, benzene 2, balance nitrogen. If the gas is burnt in a furnace with 20% excess air, calculate:
- The amount of air required per 100 kmol of gas
 - The amount of flue gas produced per 100 kmol of gas
 - The composition of the flue gases, on a dry basis
- Assume complete combustion.
- 2.5. The off-gases from a gasoline stabilizer are fed to a steam reforming plant to produce hydrogen. The composition of the off-gas, molar%, is CH_4 77.5, C_2H_6 9.5, C_3H_8 8.5, C_4H_{10} 4.5. The gases entering the reformer are at a pressure of 2 bara and 35 °C and the feed rate is 2000 m³/h.

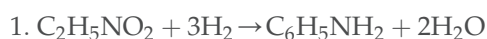
The reactions in the reformer are:



The molar conversion of $\text{C}_2\text{H}_{2n+2}$ in reaction (1) is 96% and of CO in reaction (2) is 92%. Calculate:

- a. The average molecular mass of the off-gas
- b. The mass of gas fed to the reformer, kg/h
- c. The mass of hydrogen produced, kg/h

- 2.6. Allyl alcohol can be produced by the hydrolysis of allyl chloride. Together with the main product, allyl alcohol, di-allyl ether is produced as a by-product. The conversion of allyl chloride is typically 97% and the selectivity to alcohol is 90%, both on a molar basis. Assuming that there are no other significant side reactions, calculate masses of alcohol and ether produced per 1000 kg of allyl chloride fed to the reactor.
- 2.7. Aniline is produced by the hydrogenation of nitrobenzene. A small amount of cyclo-hexylamine is produced as a by-product. The reactions are:



Nitrobenzene is fed to the reactor as a vapor, with three times the stoichiometric quantity of hydrogen. The conversion of the nitrobenzene to all products is 96%, and the selectivity for aniline is 95%.

The unreacted hydrogen is separated from the reactor products and recycled to the reactor. A purge is taken from the recycle stream to maintain the inerts in the recycle stream below 5%. The fresh hydrogen feed is 99.5% pure, with the remainder being inerts. All percentages are molar.

For a feed rate of 100 kmol/h of nitrobenzene, calculate:

- a. The fresh hydrogen feed
- b. The purge rate required
- c. The composition of the reactor outlet stream

- 2.8. Guaifenesin [guaiacol glyceryl ether, 3-(2-methoxyphenoxy)-1,2-propanediol, $\text{C}_{10}\text{H}_{14}\text{O}_4$] is an expectorant that is found in cough medicines such as Actifed and Robitussin. U.S. Patent 4,390,732 (to Degussa) describes a preparation of the active pharmaceutical ingredient (API) from guaicol (2-methoxyphenol, $\text{C}_7\text{H}_8\text{O}_2$) and glycidol (3-hydroxy propylene oxide, $\text{C}_3\text{H}_6\text{O}_2$). When the reaction is catalyzed by NaOH, the reaction yield is 93.8%. The product is purified in a thin-film evaporator, giving an overall plant yield of 87%.

- a. Estimate the feed flow rates of glycidol and guaicol that would be needed to produce 100 kg/day of the API.
- b. Estimate how much product is lost in the thin-film evaporator.
- c. How would you recover the product lost in the evaporator?

- 2.9. 11-[N-ethoxycarbonyl-4-piperidylidene]-8-chloro-6,11-dihydro-5H-benzo-[5,6]-cyclohepta-[1,2-b]-pyridine ($\text{C}_{22}\text{H}_{23}\text{ClN}_2\text{O}_2$) is a non-sedative antihistamine known as loratadine and marketed as Claritin. The preparation of the API is described in U.S. Patent 4,282,233 (to Schering). The patent describes reacting 16.2 g of 11-[N-methyl-4-piperidylidene]-8-chloro-6,11-dihydro-5H-benzo-[5,6]-cyclohepta-[1,2-b]-pyridine ($\text{C}_{20}\text{H}_{21}\text{ClN}_2$) in 200 mL of benzene with 10.9 g of ethylchloroformate ($\text{C}_3\text{H}_5\text{ClO}_2$) for 18 hours. The mixture is cooled, poured into ice water, and separated into aqueous and organic phases. The organic layer is washed with water and evaporated to dryness. The residue is triturated (ground to a fine powder) with petroleum ether and recrystallized from isopropyl ether.

- a. What is the reaction by-product?
- b. The reaction appears to be carried out under conditions that maximize both selectivity and conversion (long time at low temperature), as might be expected given the cost of the raw material. If the conversion is 99.9% and the selectivity for the desired ethoxycarbonyl substituted compound is 100%, how much excess ethylchloroformate remains at the end of the reaction?

- c. What fraction of the ethylchloroformate feed is lost to waste products?
 - d. Assuming that the volumes of water and isopropyl ether used in the quenching, washing, and recrystallization steps are the same as the initial solvent volume and that none of these materials are reused in the process, estimate the total mass of waste material produced per kg of the API.
 - e. If the recovery (plant yield) of the API from the washing and recrystallization steps is 92%, estimate the feed flow rates of 11-[N-methyl-4-piperidylidene]-8-chloro-6,11-dihydro-5H-benzo-[5,6]-cyclohepta-[1,2-b]-pyridine and ethylchloroformate required to produce a batch of 10 kg of the API.
 - f. How much API could be produced per batch in a 3.8 m³ (1000 U.S. gal) reactor?
 - g. What would be the advantages and disadvantages of carrying out the other process steps in the same vessel?
 - h. Sketch a block flow diagram of the process.
- 2.10. Describe the main commercial process used to make each of the following compounds. Include a block flow diagram.
- a. Phosphoric acid
 - b. Adipic acid
 - c. Polyethylene terephthalate
 - d. Insulin
 - e. Sorbitol
- 2.11. Example 2.7 introduced a heat-exchange system revamp problem. If the plate heat exchanger E101 in Fig. 2.19 is the gasketed-plate type, then the exchanger area can be increased by adding plates to the exchanger. How much additional area must be added to E101 to allow the system to operate at 50% above base case flow rate with no changes in temperature of steam or cooling water feed temperatures and no modification to exchangers E102 and E103?
- 2.12. Styrene is made by dehydrogenation of ethylbenzene, which is formed by alkylation of benzene with ethylene. Propose target yields for the alkylation and dehydrogenation steps if the forecasted prices per metric ton are styrene \$800, ethylene \$800, and benzene \$500.
- Note: Structures for the compounds in Problems 2.3, 2.8, and 2.9 can be found in the Merck Index, but are not required to solve the problems.